UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Unit 1 · Partial Differentiation · all sessions (Class 001 – Class 011) · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript “Introduction to Partial Differentiation”, segment start
T001b 00:11:41	Transcript “Continuity and differentiability of a function of two independent variables”
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question
S003·7 · S004·9	Deck 003 / 004, slide number (rendered .pptx)
T003a 00:06:27	Transcript 003 part I: “Total derivative, chain rule and implicit functions”
T003b 00:01:10	Transcript 003 part II: “Problems on total derivatives, chain rule”
T004a 00:13:09	Transcript 004 part I: “Implicit functions and problems”
T004b 00:02:31	Transcript 004 part II: “Problems on composite and implicit functions”
CW U1 P14 · ESA Jan-24 s3 1b	Unit 1 class-work problem 14 · past-paper question
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)

S007·9	Deck 007, slide 9 (rendered .pptx)
T007 00:18:07	AV-summary transcript 007, segment start
CW U1 P24	Unit 1 class-work problem 24
ESA Dec-23 1b	Past paper: Dec 2023, question 1.b
S008·12	Deck 008, slide 12 (rendered .pptx)
T008 00:07:38	AV-summary transcript 008, segment start
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c
S010·8	Deck 010, slide 8 (rendered .pptx)
T010a 00:07:38	AV-summary transcript “Extremas of functions continued” (older lecture), segment start
T010b 00:03:12	AV-summary transcript “Problems on Maxima and Minima for a function of two variables” (newer lecture)
CW U1 P29	Unit 1 class-work problem 29
ESA Jan-24 s2 1a	Past paper: Jan 2024 set 2, question 1.a
S011·5	Deck 011, slide 5 (rendered .pptx)
T011a 00:01:08	Transcript “Lagrange's method of undetermined coefficient” (theory lecture), segment start
T011b 00:05:03	Transcript “Problems on Lagrange's method of undetermined multipliers”
CW U1 P33	Unit 1 class-work problem 33
Extra Q57	Extra question list “lagrange-ques”, question 57
ESA Jul-23 1c	Past paper: Jul 2023, question 1.c

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Class 001. Colours follow the Okabe–Ito order: blue = main surface or curve, vermilion = comparison, green and purple = further paths. Figures G1.1–G1.2 and G3.1–G3.2 redraw the deck's Gnuplot surfaces ($\sin x \cos y$, $e^{1-x^2-y^2}$, $\frac{3x^2y}{x^2+y^2}$, $\frac{2xy}{3x^2+y^2}$) over $-2 \le x, y \le 2$ or a similar range; the slides' screenshots are not copied. Figure G2.1 redraws the lecture's board sketch of paths approaching (a, b).

Class 002. Colours follow the Okabe–Ito order: blue = main curve, vermilion =

comparison or tangent, green = second trace, purple = other direction. Tangents are dashed. Grey dashed lines are references. The surfaces (dome $z = 4 - x^2 - y^2$, saddle $z = x^2 - y^2$, $z = x^2 y$) are illustrative choices that reproduce the shapes on slides 5–10. The slide images themselves are third-party screenshots and are not copied.

Classes 003–004. Colours follow the Okabe–Ito order: blue = main curve or first branch, vermilion = tangent, comparison or second branch, green = third series, purple = other directions. Dashed lines are tangents or approximations; grey dashed lines are references. Tree diagrams are redrawn from slides 003·5, 003·14 and 004·5. The 3-D picture of the point moving on the saddle and cylinder is drawn from the equations of slide 003·7.

Class 005. Colours follow the Okabe–Ito order. Both figures are added for clarity, since the deck has no graphs: one shows how a homogeneous function scales along a ray, the other shows level curves of $x^2 + xy + y^2$ as scaled copies.

Class 006. Colours follow the Okabe–Ito order: blue, vermilion, green, purple for the four deck functions in G1.1; grey dashed lines are references. The deck has no graphs, so both figures are added for clarity and drawn from the deck's own functions.

Class 007. Colours follow the Okabe–Ito order: blue = the function, vermilion = first approximation, green and purple for higher approximations; grey dashed lines are references. Figure G1.1 redraws the slide's graphs of $\sin x$ and its Maclaurin polynomials; the slide's own axis scale is not stated, so the redraw uses $-2\pi \le x \le 2\pi$. Figures G3.1–G3.2 redraw the deck's surface plots of $xe^y + 1$; the slides' screenshots are not copied.

Class 008. Colours follow the Okabe–Ito order: blue = the function, vermilion = first-degree polynomial, green = second degree, purple = third degree; grey dashed lines are references. The deck has no graphs, so all figures are added for clarity and drawn from the deck's own functions.

Class 009. Colours follow the Okabe–Ito order: blue = the surface or curve, vermilion = the stationary point and its tangent plane, green = second trace. Figures G1.1–G1.3 redraw the deck's pictures of maxima, minima and a saddle with the illustrative functions $e^{-x^2-y^2}$-type bumps, $x^2 + y^2$ and $x^2 - y^2$; the slides' screenshots (including a photograph of a horse saddle) are not copied.

Class 010. The deck has no graphs, so every figure is added for clarity and drawn from the deck's and the papers' own functions. Contour plots: thin grey lines are level curves; a filled vermilion triangle marks a maximum, a filled blue circle a minimum and a green cross a saddle point, each labelled directly.

Class 011. Colours follow the Okabe–Ito order: blue = the constraint curve, vermilion and green = level curves of the function being optimised, grey = guides. Figure G1.1 is added for clarity (why the method works). Figure G2.1 redraws the deck's "soap bubble" picture of slide 011·7 as a cross-section; Figure G2.2 draws the ESA curve $5x^2 + 6xy + 5y^2 = 8$ with its nearest and farthest points.

M1

Class 001 · Introduction to Partial Differentiation

1 Functions of two or more variables

$$z = \sin x \cos y$$

$$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$$

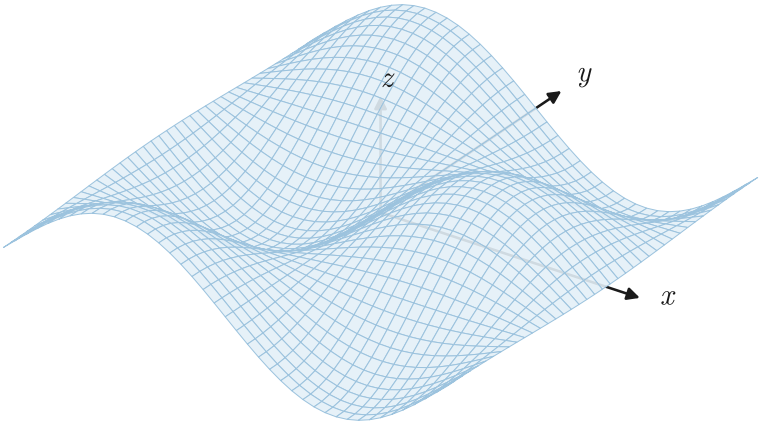


Figure G1.1. Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A function of two variables is a surface over the xy-plane: above each point (x, y) the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between -1 and 1; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

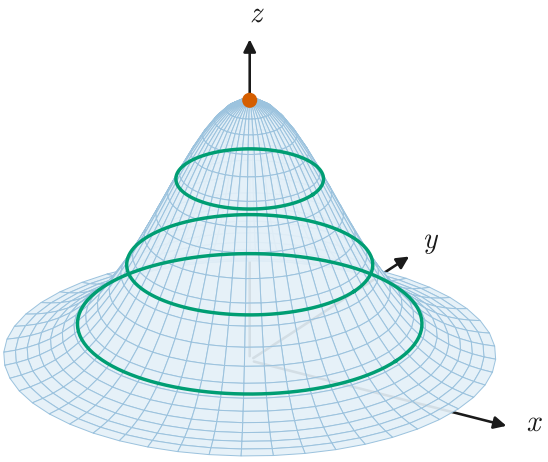


Figure G1.2. Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

How to read it. A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

2 Limit of a function of two variables

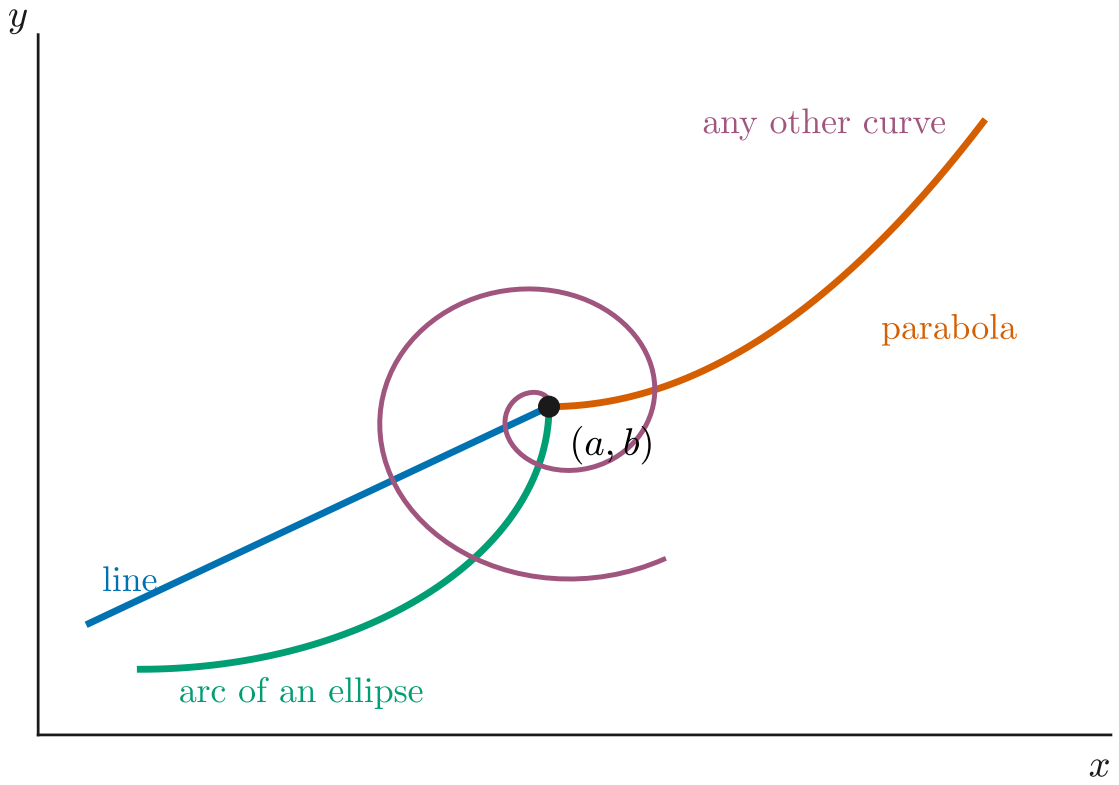


Figure G2.1. Paths along which (x, y) can approach (a, b). ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

How to read it. In one variable x reaches a only from the left or the right. In the plane, (x, y) can reach (a, b) along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

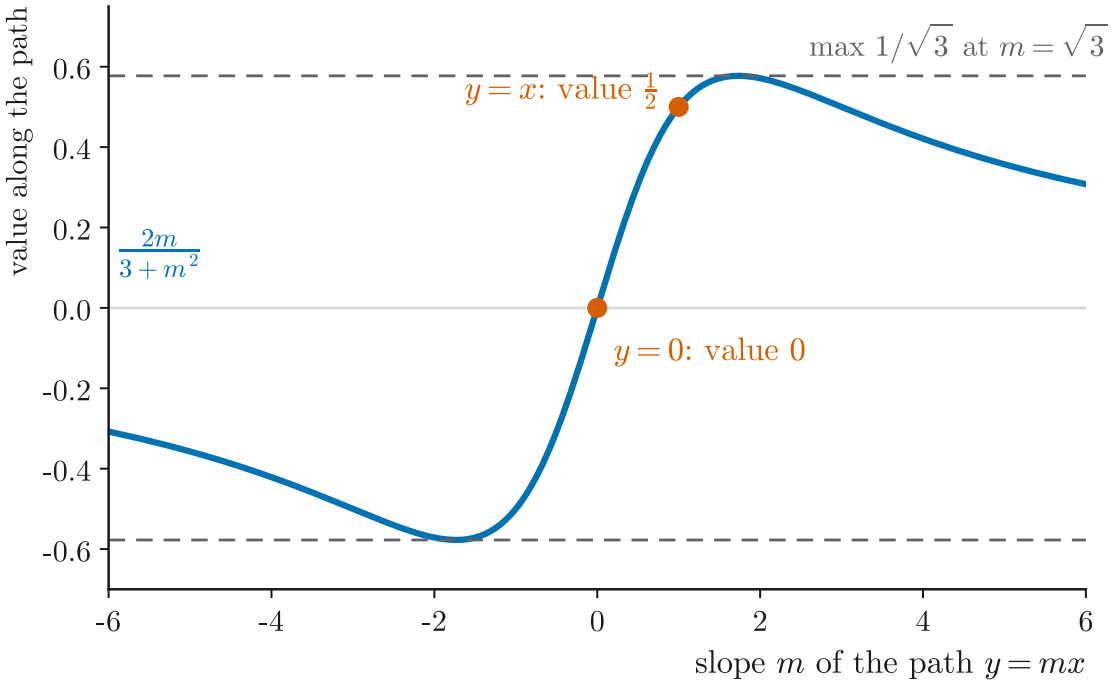


Figure G2.2. $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

How to read it. The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope m. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

3 Continuity of a function of two variables

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

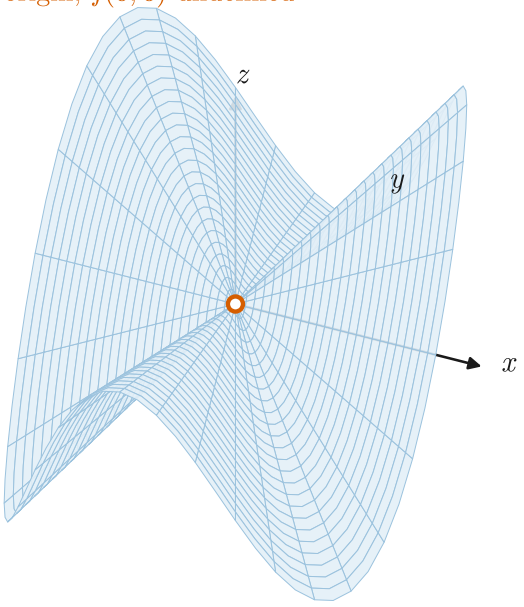


Figure G3.1. $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

How to read it. The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where f is undefined: a removable discontinuity.

$$z = \frac{2xy}{3x^2 + y^2}$$

ray $y = x$: height $\frac{1}{2}$

x-axis: height 0

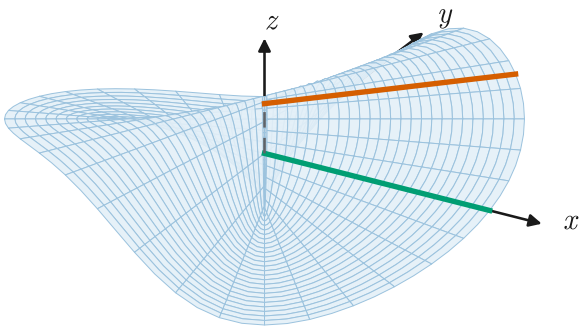


Figure G3.2. $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

How to read it. Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the z-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the x-axis at height 0). There is no single value at the origin.

M2

Class 002 · Geometrical Interpretation of Partial Derivatives

7 The one-variable picture: derivative as slope

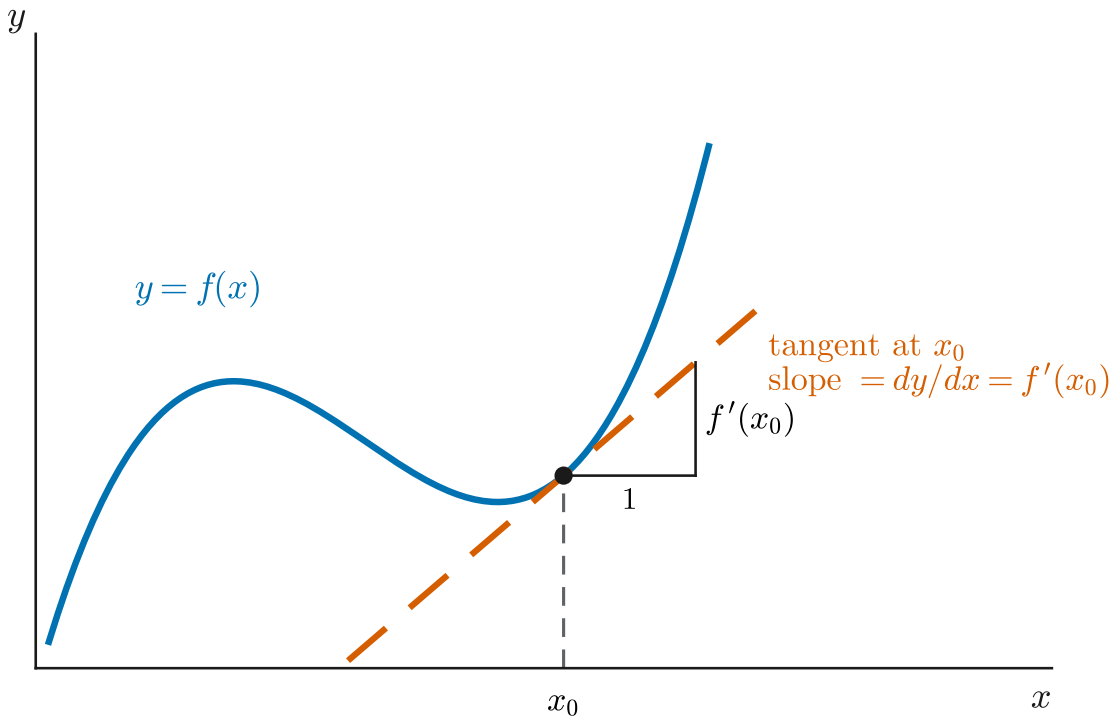


Figure G7.1. Derivative of y = f(x) as the slope of the tangent. ESA ★

S002·4 · T002 00:03:13

How to read it. The blue curve is $y = f(x)$, and the dashed vermilion line is its tangent at x_0. The small triangle shows the slope: a run of 1 gives a rise of $f'(x_0)$. In an exam, label the point, the tangent and the slope dy/dx at x_0.

8 A function of two variables is a surface

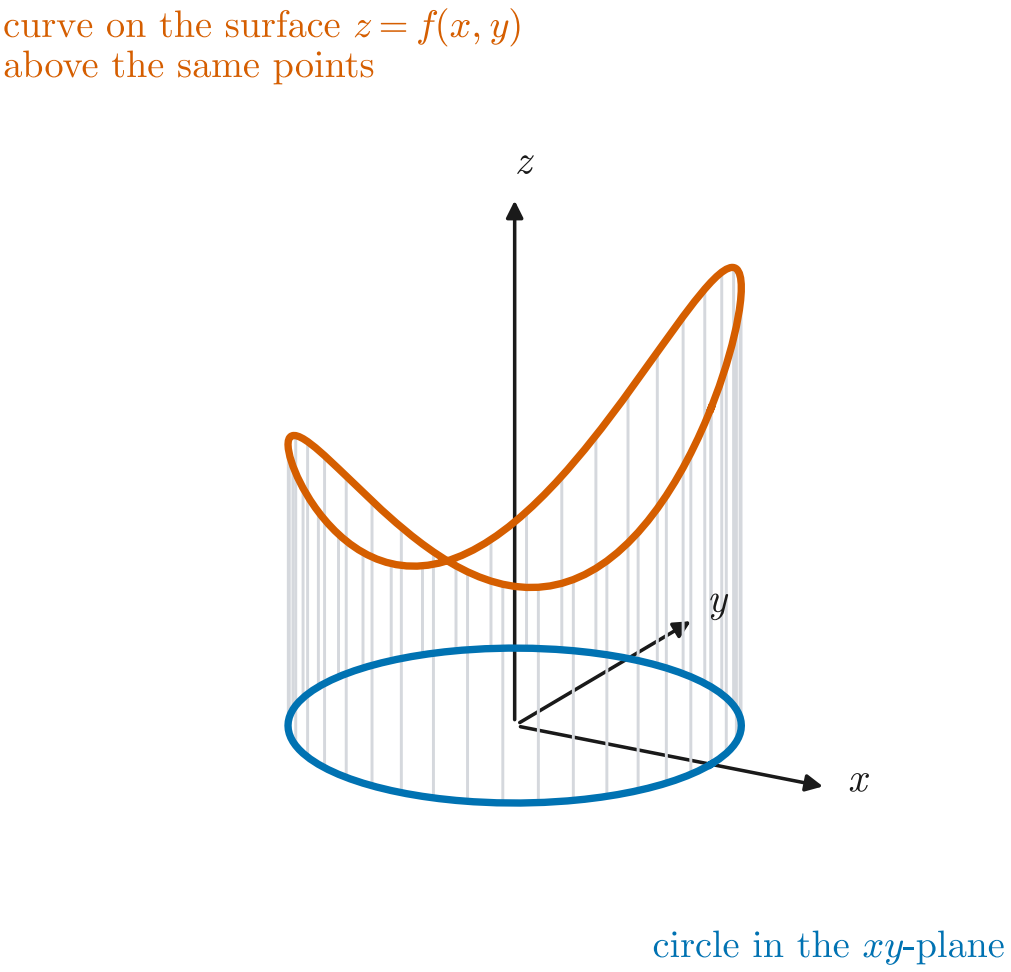


Figure G8.1. Points on a circle in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:32

How to read it. The blue circle lies in the xy-plane. Each grey line rises to $z = f(x, y)$, and the vermilion curve joins the tops: a closed curve lying on the surface $z = f(x, y) = 2.2 + 0.8xy + 0.5x$ (an illustrative choice).

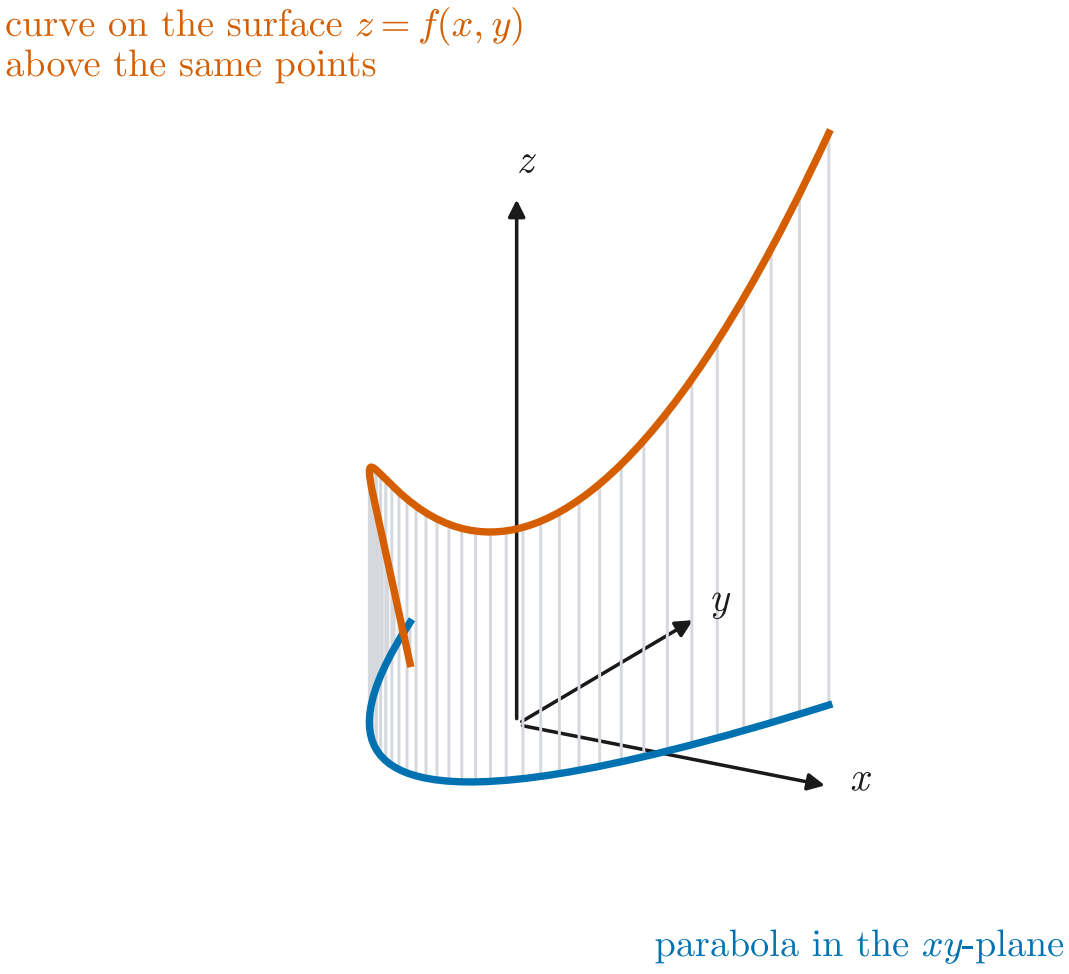


Figure G8.2. Points on a parabola in the xy-plane give a curve on the surface. ESA ★

S002·5 · T002 00:03:47

How to read it. Same idea with a parabola $y = x^2 - 1.2$ in the xy-plane. Every point of the parabola has one height on the surface, so the vermilion curve lies on the surface above it.

9 First-order partials as slopes of tangents

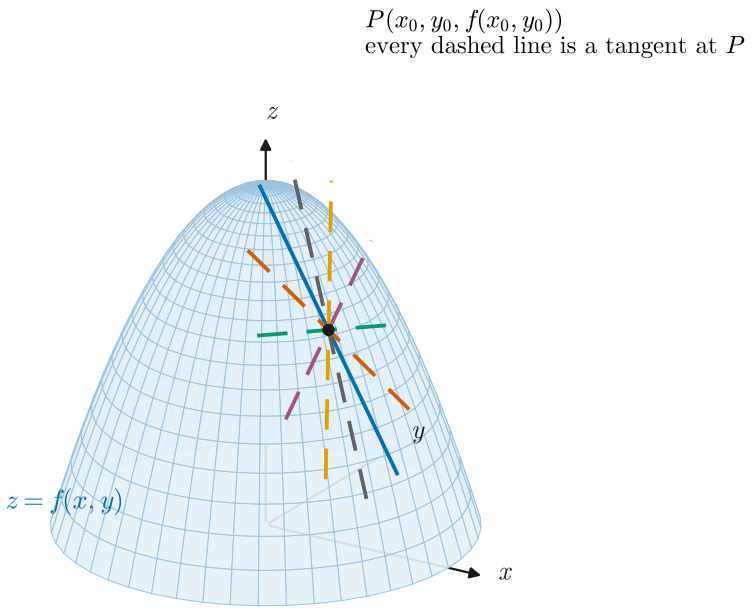


Figure G9.1. A surface has infinitely many tangents at a point. ESA ★★

S002·6 · T002 00:04:36 · T002 00:05:24

How to read it. The dome is $z = 4 - x^2 - y^2$ and $P = (1, -0.5, 2.75)$. Each coloured line is a tangent to the surface at P in a different direction. Partial differentiation picks out two of them: the ones parallel to the x- and y-axes.

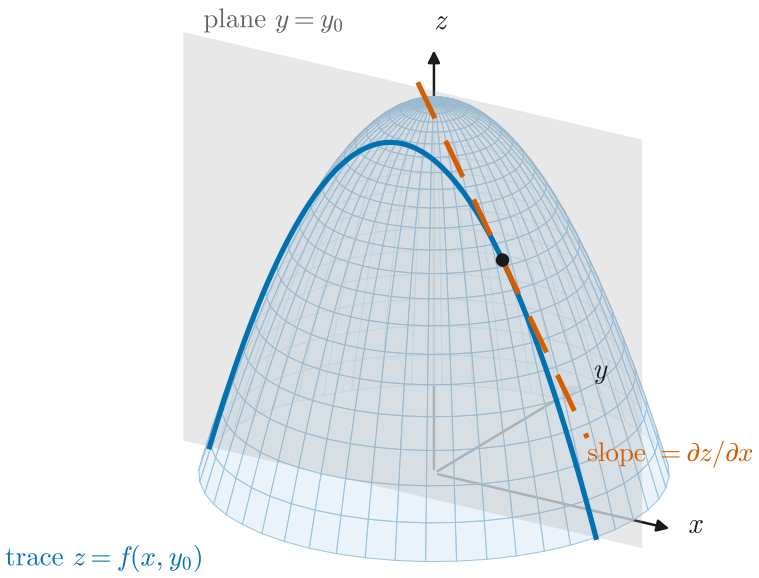


Figure G9.2. $\partial z/\partial x$: the tangent parallel to the x-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:06:54

How to read it. The grey plane $y = y_0$ cuts the dome in the blue trace $z = f(x, y_0)$. The dashed tangent to that trace at P is parallel to the x-axis, and its slope is $\partial z/\partial x$ at (x_0, y_0).

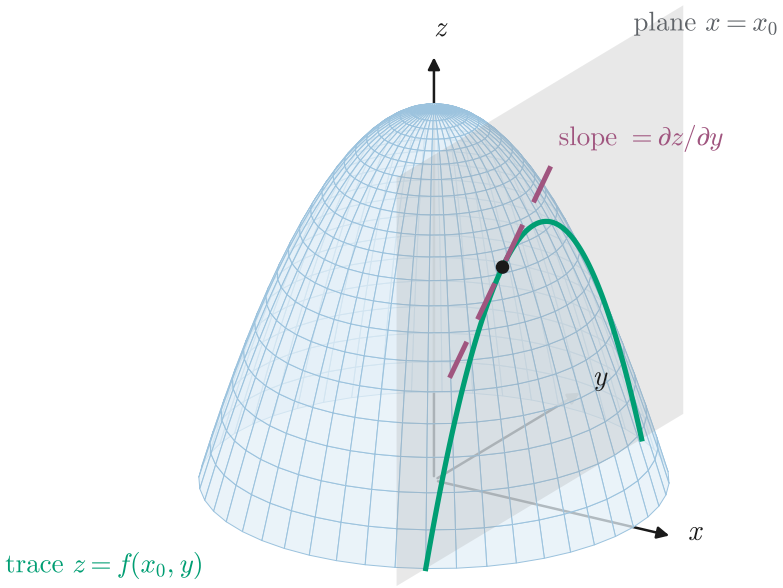


Figure G9.3. $\partial z/\partial y$: the tangent parallel to the y-axis. ESA ★★

S002·7 · V002 00:07:45 · T002 00:07:10

How to read it. The plane $x = x_0$ cuts the dome in the green trace $z = f(x_0, y)$, which "just depends on y". The dashed tangent at P is parallel to the y-axis, and its slope is $\partial z/\partial y$ at (x_0, y_0).

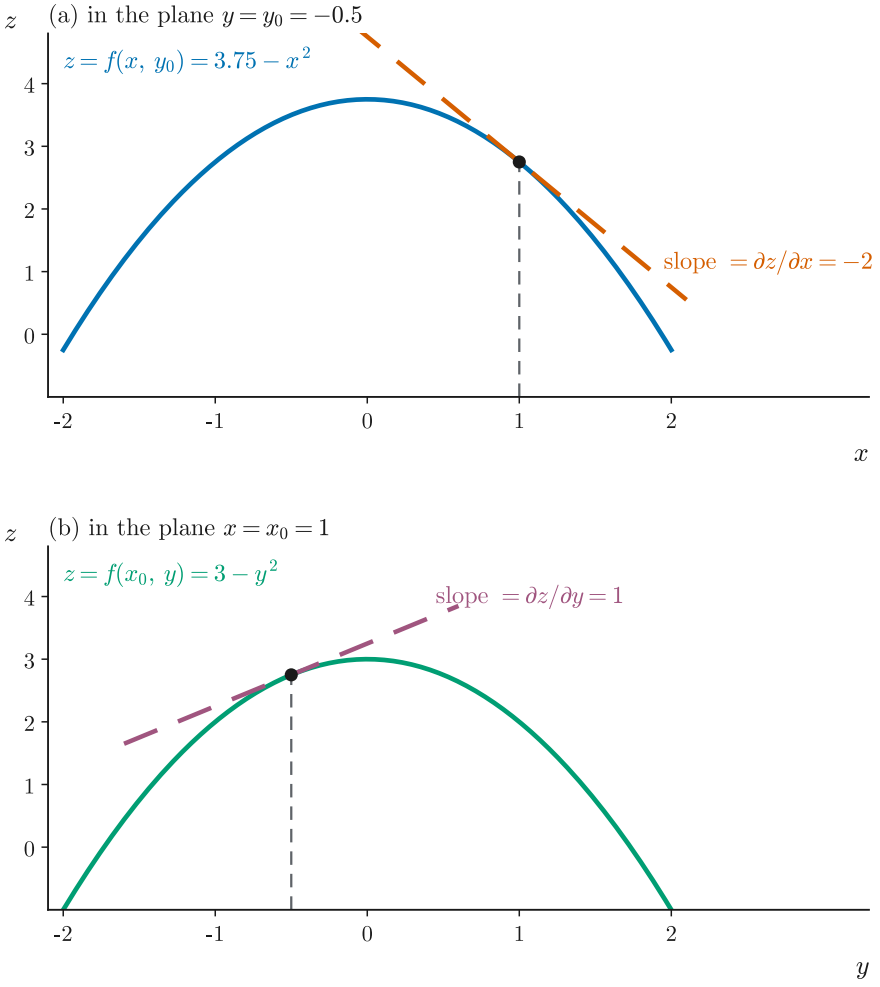


Figure G9.4. The two traces seen flat, with numerical slopes. ESA ★★

ADDED FOR CLARITY S002·7

How to read it. These are panels (a) and (b) from Figures G9.2 and G9.3, drawn in their own planes. At P, the tangent in (a) has slope $-2 = f_x(1, -0.5)$ and the tangent in (b) has slope $1 = f_y(1, -0.5)$. See worked example W9.1.

10 Slope in any other direction: the gradient

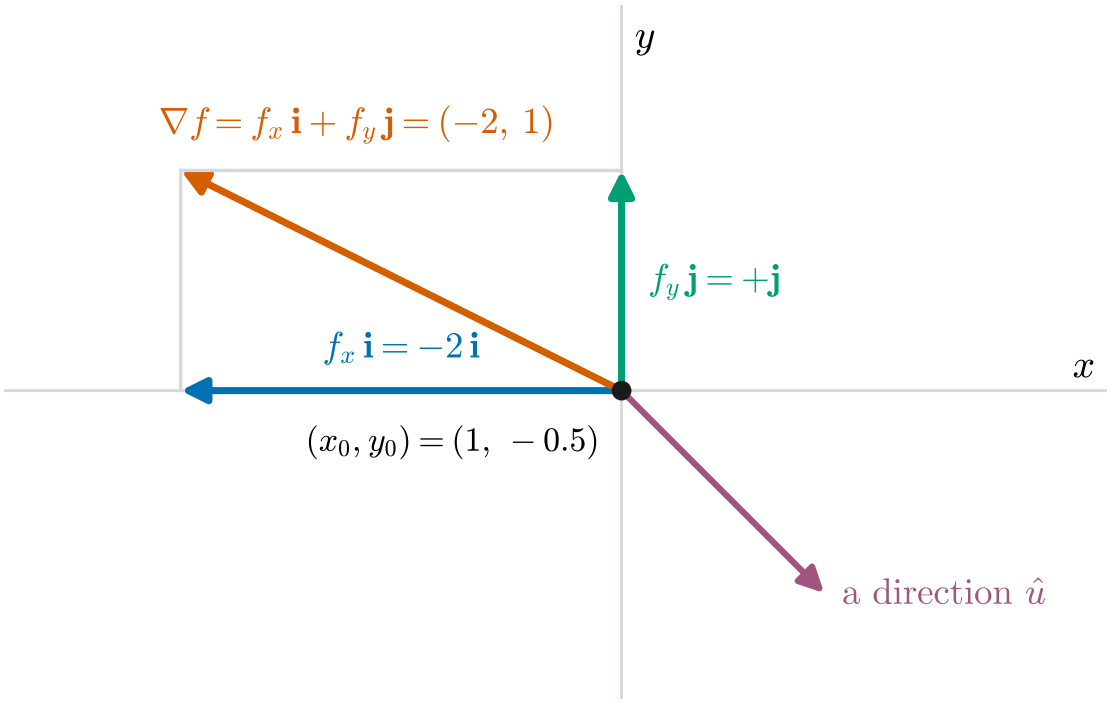


Figure G10.1. Resolving into components: the gradient. ESA ★

S002·8 · T002 00:09:06 · V002 00:09:05

How to read it. The slope components $f_x\,\mathbf{i}$ (blue) and $f_y\,\mathbf{j}$ (green) add by vector addition to give the resultant ∇f (vermilion), drawn here for the dome at $(1, -0.5)$. The purple arrow is another direction $\hat{u}$; the slope along it is $\nabla f \cdot \hat{u}$ (W10.1).

11 Second-order partials: concavity and mixed partials

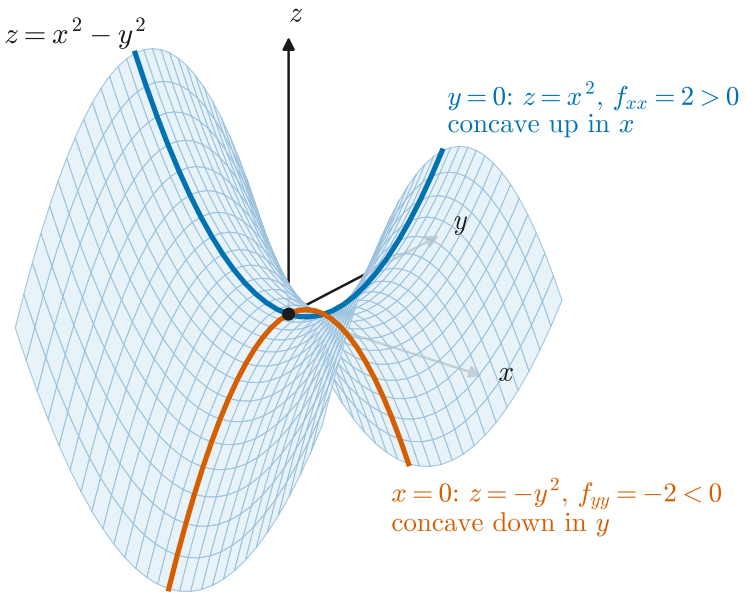


Figure G11.1. Concave up in x, concave down in y: the saddle z = x² − y². ESA ★★

ADDED FOR CLARITY S002·9 · S002·10 · T002 00:11:19

How to read it. Along the x direction (blue trace, $y = 0$) the surface bends up, so $f_{xx} = 2 > 0$. Along the y direction (vermilion trace, $x = 0$) it bends down, so $f_{yy} = -2 < 0$. One point can be concave up in one direction and concave down in another.

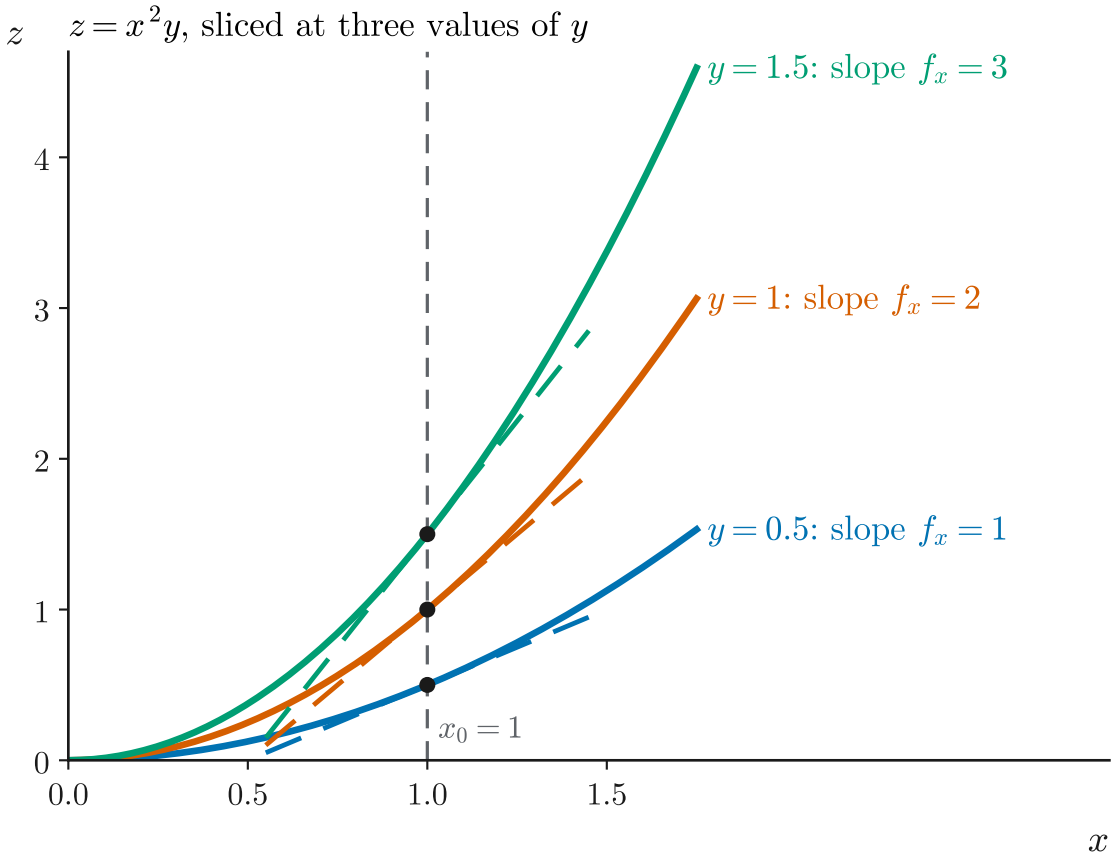


Figure G11.2. Mixed partial: how the x-slope changes as y changes. ESA ★★

ADDED FOR CLARITY S002·9 · T002 00:11:45

How to read it. Three traces of $z = x^2 y$, at $y = 0.5$, 1 and 1.5. At $x_0 = 1$ their tangents have slopes $f_x = 1, 2, 3$. The x-slope grows by 2 per unit of y, which is $f_{xy} = 2$.

M3

Classes 003–004 · Total Derivative, Chain Rule and Implicit Functions

14 The total derivative rule

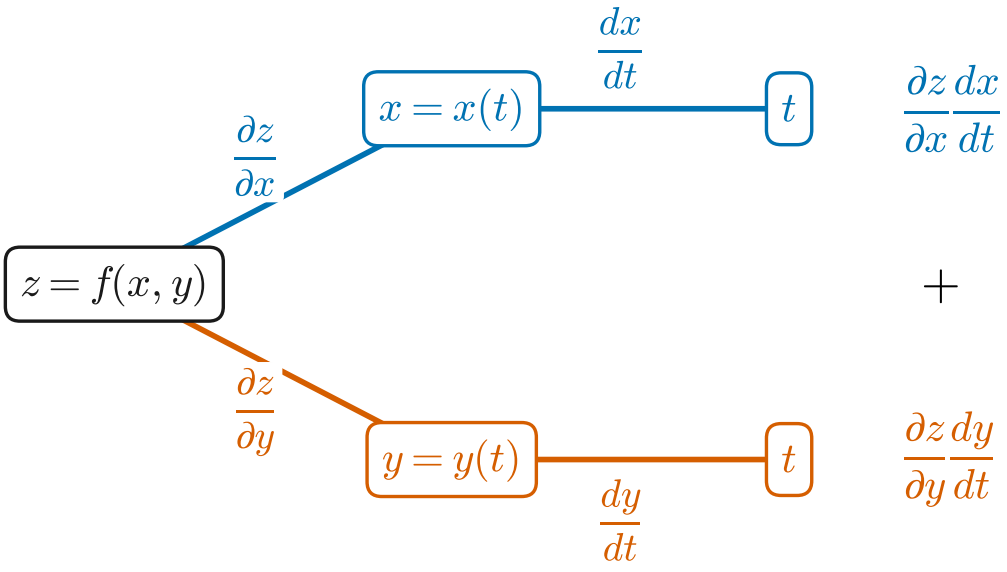


Figure G14.1. Tree diagram for the total derivative. ESA ★★

S003·5 · T003a 00:19:22

How to read it. Start at z. Each branch carries the derivative of the upper variable with respect to the lower one. Multiply along each path from z down to t, then add the two paths: $\dfrac{dz}{dt} = \dfrac{\partial z}{\partial x}\dfrac{dx}{dt} + \dfrac{\partial z}{\partial y}\dfrac{dy}{dt}$. In an exam, draw it first so that no path is missed.

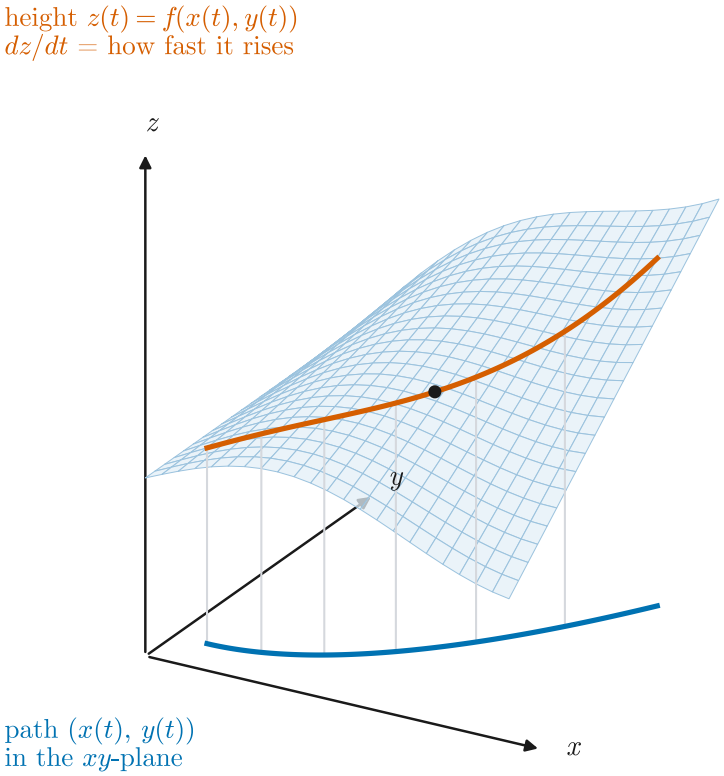


Figure G14.2. The total derivative as the slope along a path. ESA ★

ADDED FOR CLARITY S003·5 · T003a 00:05:20

How to read it. As t runs, the point $(x(t), y(t))$ moves along the blue path in the xy-plane. The vermilion curve is the surface height above it, $z(t) = f(x(t), y(t))$. dz/dt is how fast that height changes as t increases; it combines the slopes z_x and z_y weighted by how fast x and y move. The surface is an illustrative choice.

15 The total differential: approximation and sensitivity

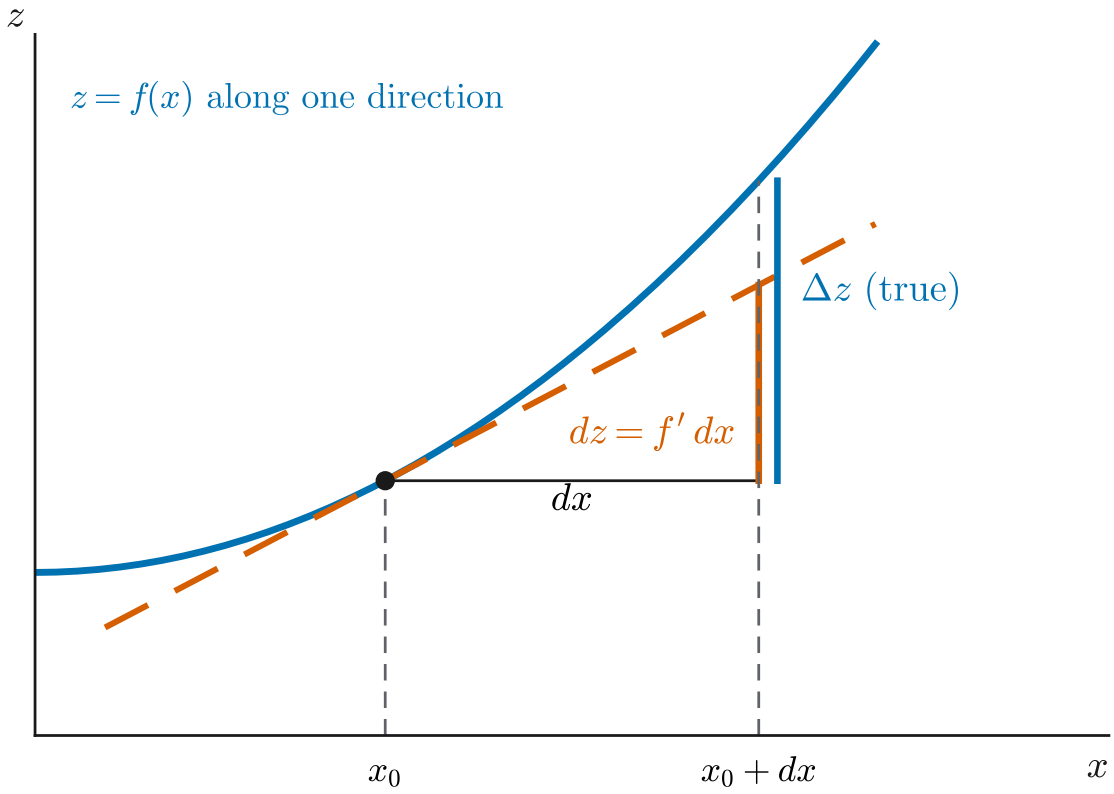


Figure G15.1. The differential approximates the true change. ESA ★★

ADDED FOR CLARITY S003·8 · S004·4

How to read it. Cut along one direction. The true change Δz follows the blue curve; the differential dz follows the dashed tangent line. For a small step dx the two are nearly equal, which is why $f(x_0 + dx) \approx f(x_0) + dz$. The gap grows as the step grows.

16 Rates of change problems

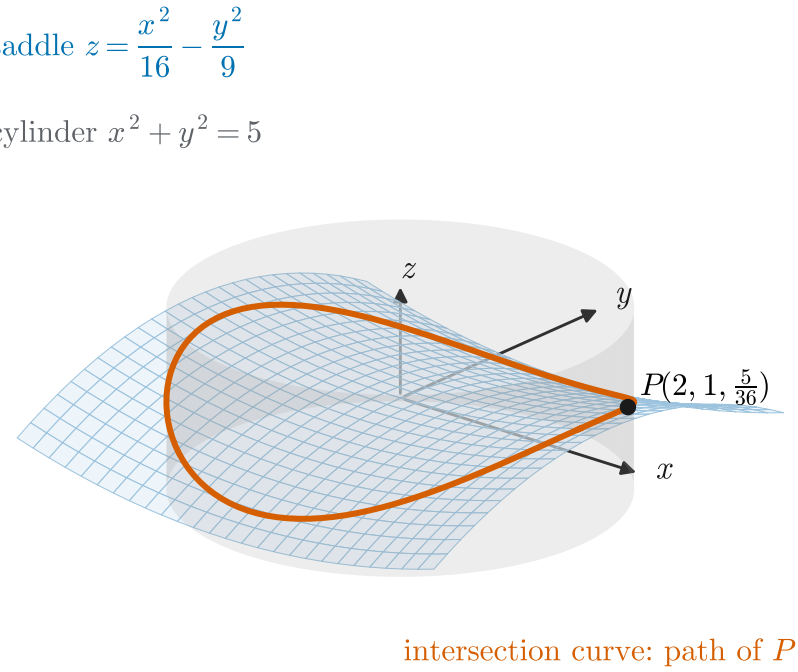


Figure G16.1. A point moving on a saddle and a cylinder. ESA ★★

S003·7 · T003a 00:06:27

How to read it. The saddle $z = x^2/16 - y^2/9$ (light blue) and the cylinder $x^2 + y^2 = 5$ (grey) meet in the vermilion curve. The point P moves along that curve; at $x = 2$, $y = 1$ its height rises at $dz/dt = 5/36$ cm/s when $dx/dt = 0.2$ cm/s (W16.1). The slide calls the surface a paraboloid, but it is a hyperbolic paraboloid (a saddle).

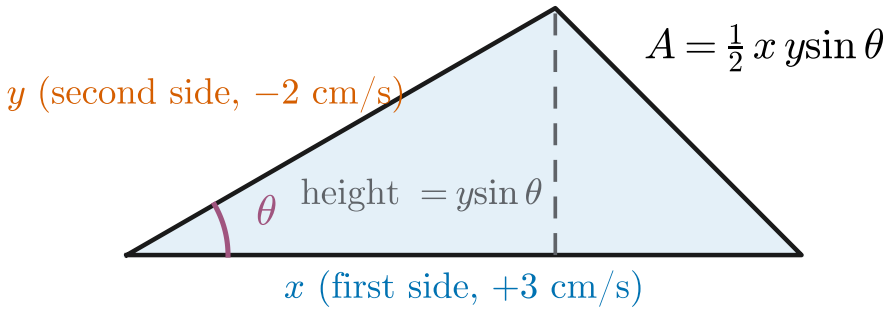


Figure G16.2. Triangle with two sides and the included angle. ESA ★★

S003·9 · T003a 00:15:25

How to read it. Sides x and y meet at the angle θ. The dashed perpendicular is the height $y \sin\theta$, so the area is $A = \frac{1}{2} x y \sin\theta$. Label the three changing quantities x, y, θ before differentiating.

17 The chain rule for composite functions

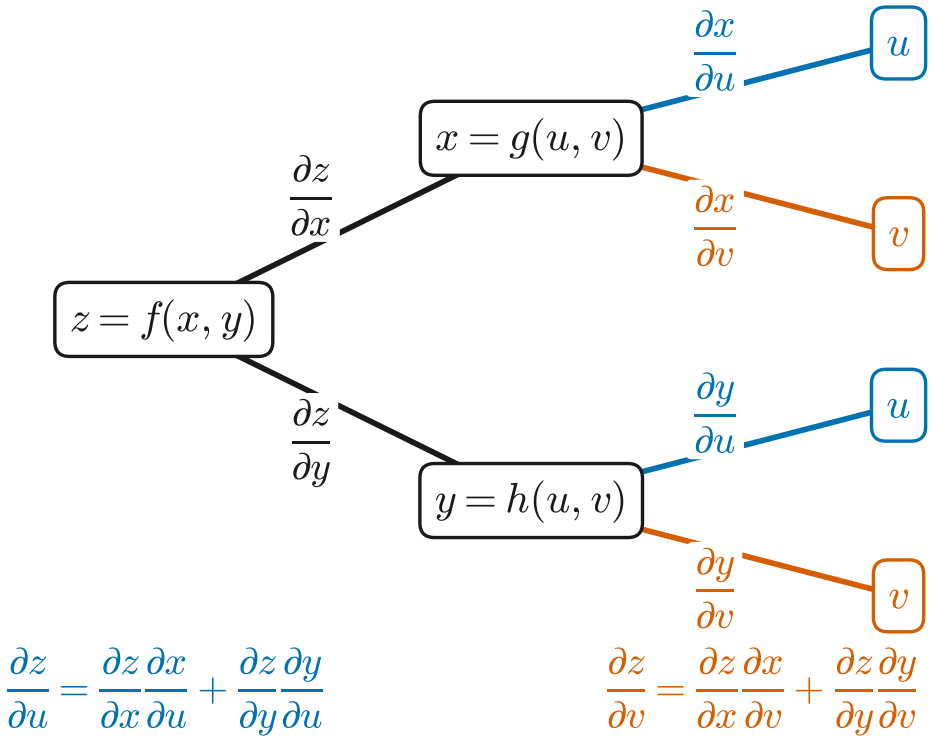


Figure G17.1. Tree diagram for the chain rule. ESA ★★★

S003·14 · T003a 00:19:42

How to read it. z branches to x and y, and each of those branches to both u and v, giving four leaves. For $\partial z/\partial u$ follow only the two paths ending at u (blue): $z_x x_u + z_y y_u$. For $\partial z/\partial v$ follow the two ending at v (vermilion). In an exam, draw the tree first and read the terms off it.

18 Implicit functions of two variables

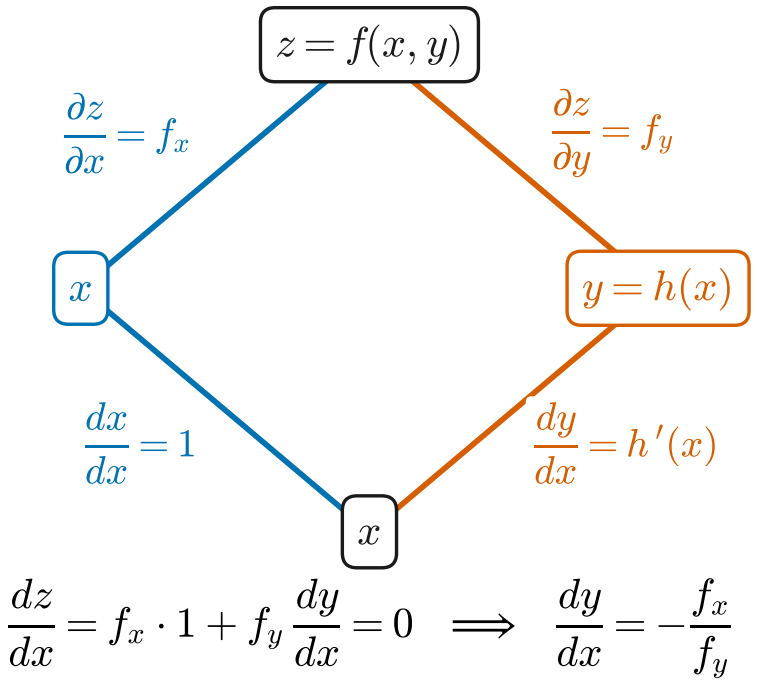


Figure G18.1. Branch diagram for implicit differentiation. ESA ★★

S004·5 · T004a 00:07:57

How to read it. Differentiate $z = f(x, y)$ with respect to x along both branches. The left branch is direct ($f_x \cdot dx/dx = f_x \cdot 1$). The right branch goes through $y = h(x)$ ($f_y \cdot dy/dx$). On a level curve the total is 0: $f_x + f_y\, dy/dx = 0$.

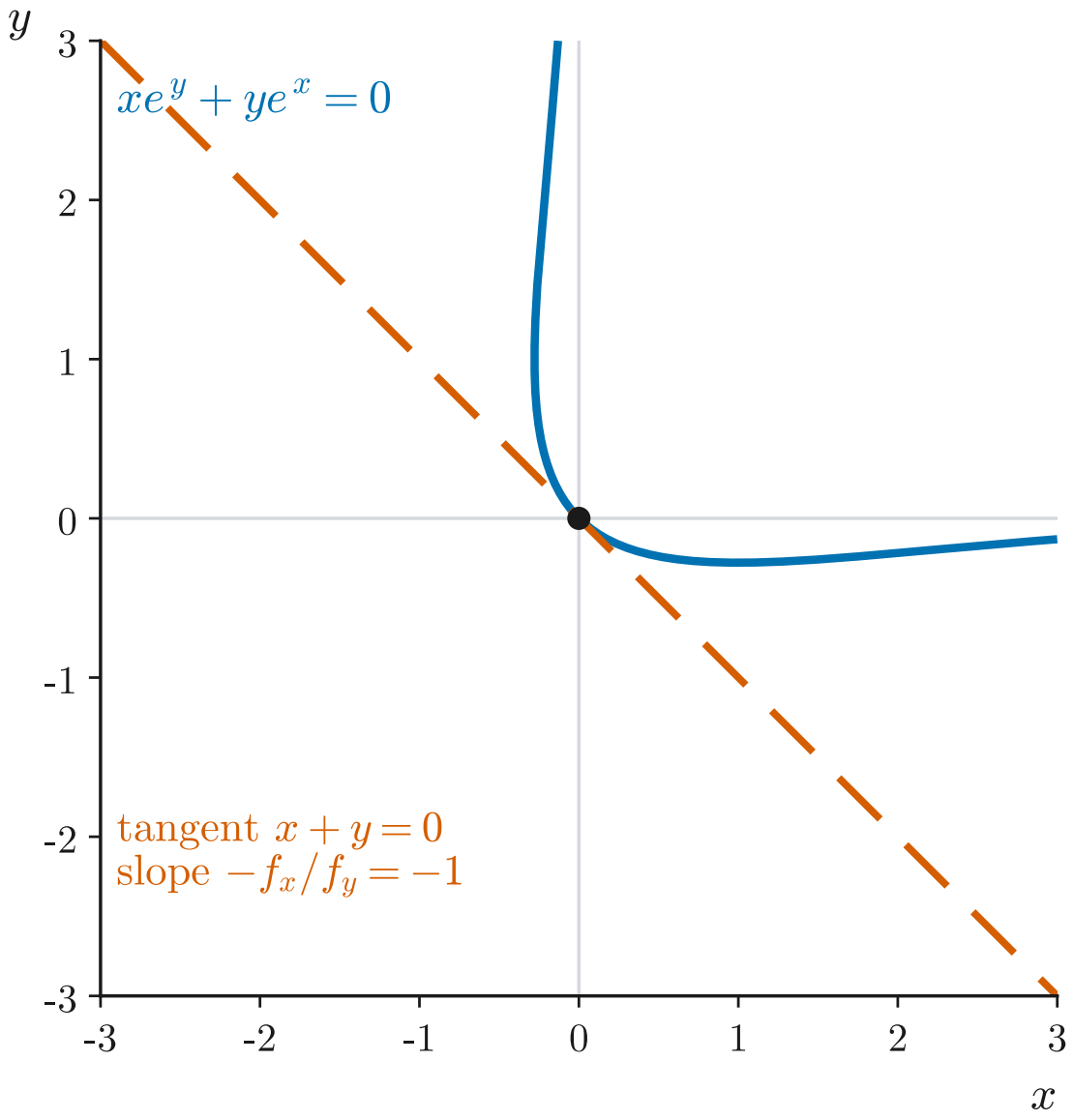


Figure G18.2. The curve $xe^y + ye^x = 0$ and its tangent at the origin. ESA ★★

S004·7 · T004a 00:13:48

How to read it. The blue curve is the set where $f(x, y) = xe^y + ye^x = 0$. At the origin $dy/dx = -f_x/f_y = -1$, so the dashed tangent is $y = -x$. The curve is plotted from its equation (the slide's picture is a 3-D grapher screenshot).

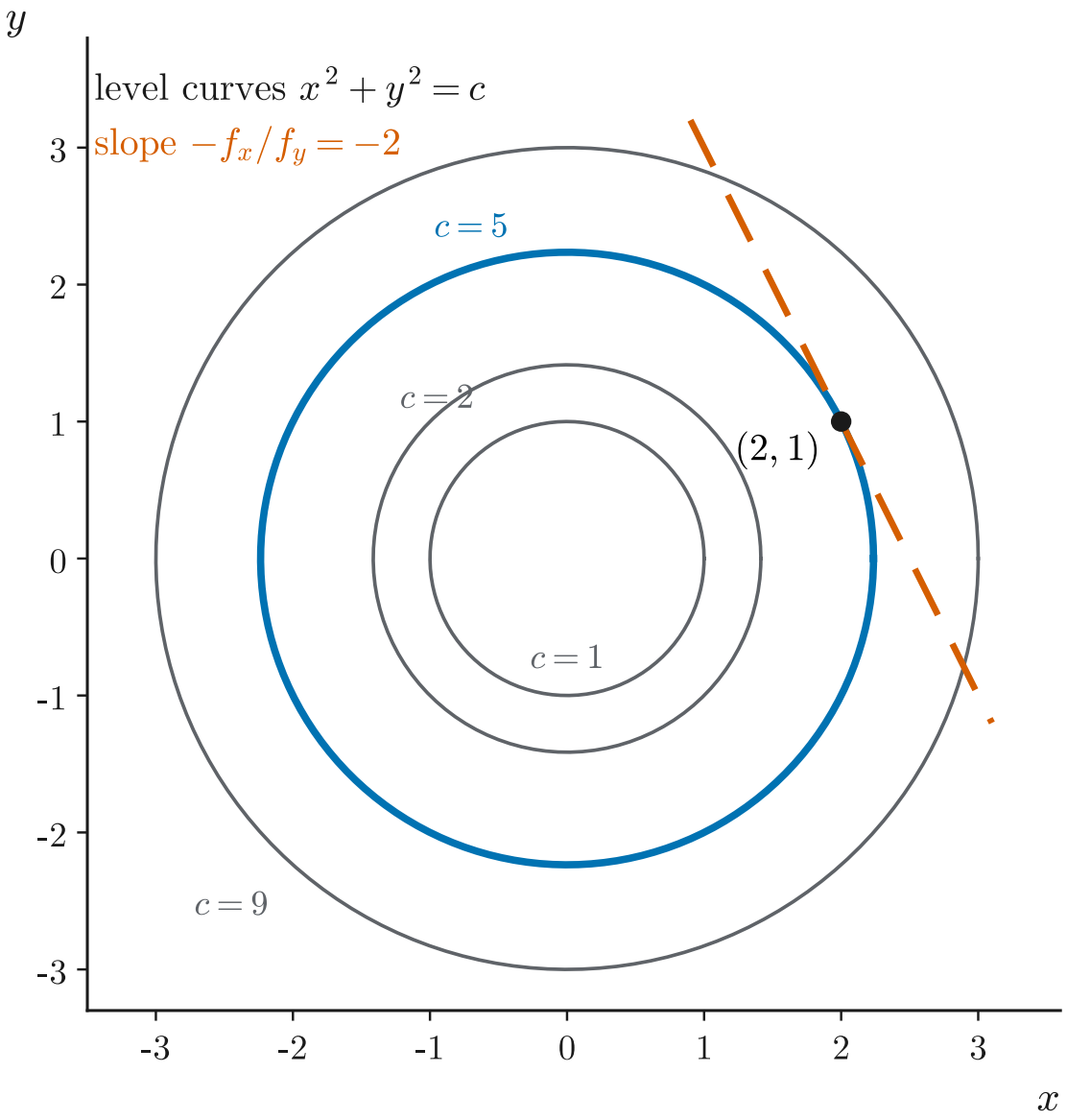


Figure G18.3. Level curves and the slope $-f_x/f_y$. ESA ★★

ADDED FOR CLARITY S004·4 · S003·7 · T004a 00:03:10

How to read it. Level curves of $f = x^2 + y^2$ for $c = 1, 2, 5, 9$: the cross-sections of the cylinders on slide 004·4. On $c = 5$ at $(2, 1)$ the dashed tangent has slope $-f_x/f_y = -2x/2y = -2$, the same $dy/dx = -x/y$ used in W16.1.

M4

Class 005 · Homogeneous Functions and Euler's Theorem

20 Homogeneous functions

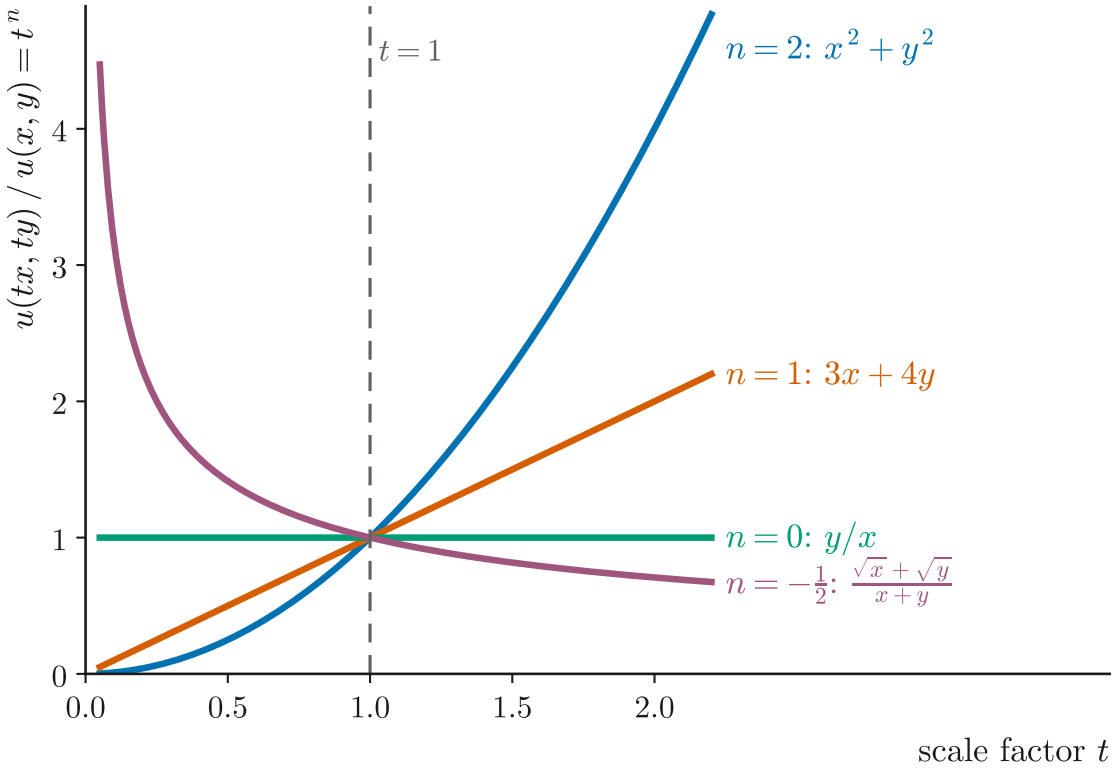


Figure G20.1. Scaling along a ray: $u(tx,ty)=t^n u(x,y)$. ESA ★★

ADDED FOR CLARITY S005·3 · S005·5

How to read it. Start at a point and move out along the ray through the origin by the factor t. A homogeneous function of degree n changes like t^n: degree 2 grows quadratically, degree 1 linearly, degree 0 stays constant along the ray (but can vary between rays), and degree $-\frac{1}{2}$ decays. Each curve is $u(t)/u(1)$.

21 Euler's theorem

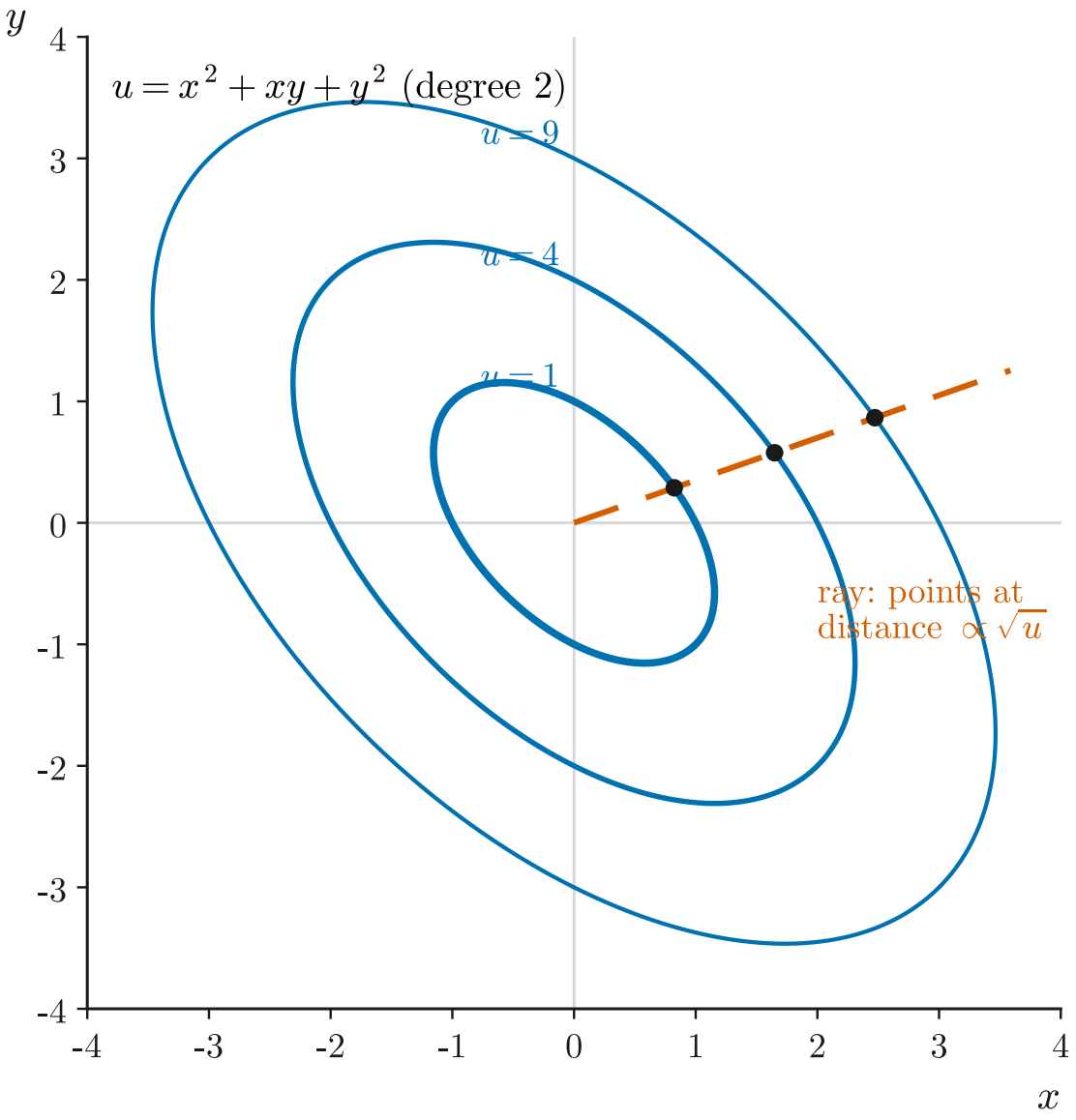


Figure G21.1. Level curves of a homogeneous function are scaled copies. ESA ★★

ADDED FOR CLARITY S005·5 · T005 00:10:18

How to read it. Level curves of $u = x^2 + xy + y^2$ (degree 2) for $u = 1, 4, 9$. Each is the first one scaled by $1, 2, 3 = \sqrt{u}$. Moving out along the dashed ray, the rate of change $xu_x + yu_y$ (the radial change scaled by distance) equals $2u$: this is Euler's theorem.

M5

Class 006 · Problems on Euler's Theorem

25 Testing homogeneity: the scaling test

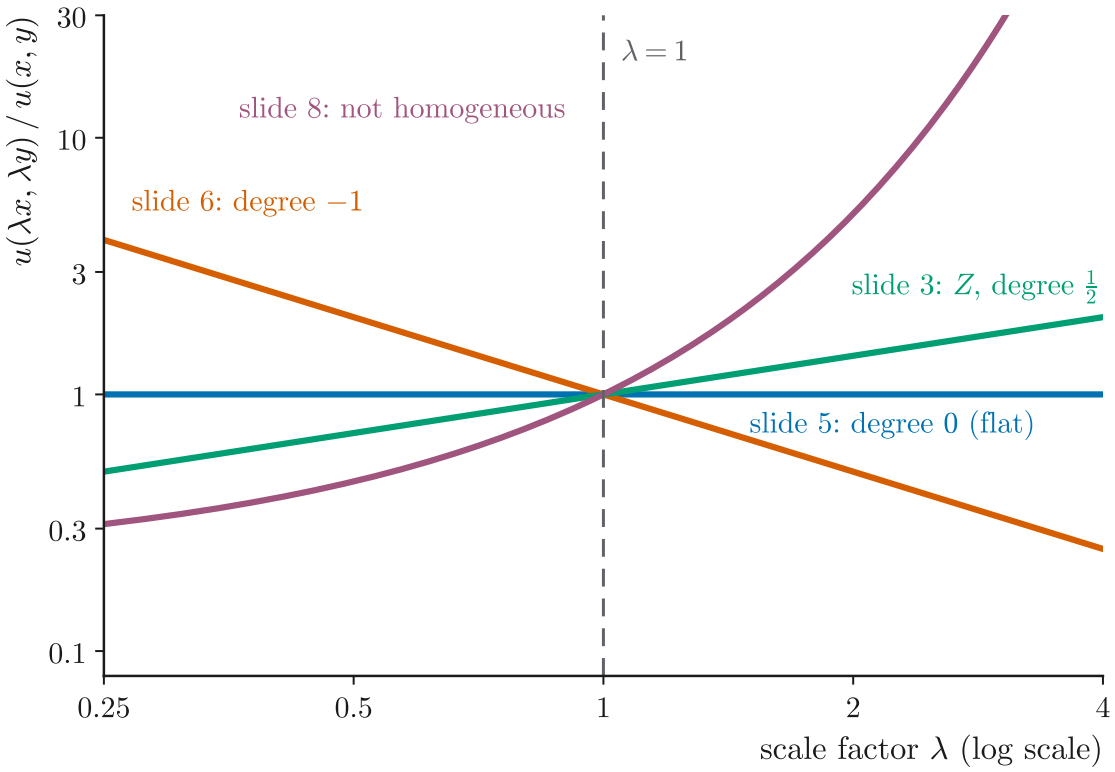


Figure G25.1. The scaling test on log-log axes. ESA ★★

ADDED FOR CLARITY S006·3 · S006·5 · S006·6 · S006·8

How to read it. Each curve is $u(\lambda x, \lambda y)/u(x, y)$ for a deck function, plotted against λ on log-log axes. A homogeneous function gives λ^n: a straight line of slope n through $(1, 1)$. Slide 006·5 is flat (degree 0), slide 006·6 falls with slope -1, Z from slide 006·3 rises with slope $\frac{1}{2}$. Slide 006·8's function curves away: it is not homogeneous. Base point $(1, 2)$; $(2, 1)$ for slide 006·6, where $\sin^{-1}(y/x)$ needs $y \leq x$.

28 When f(u) is homogeneous: the extension

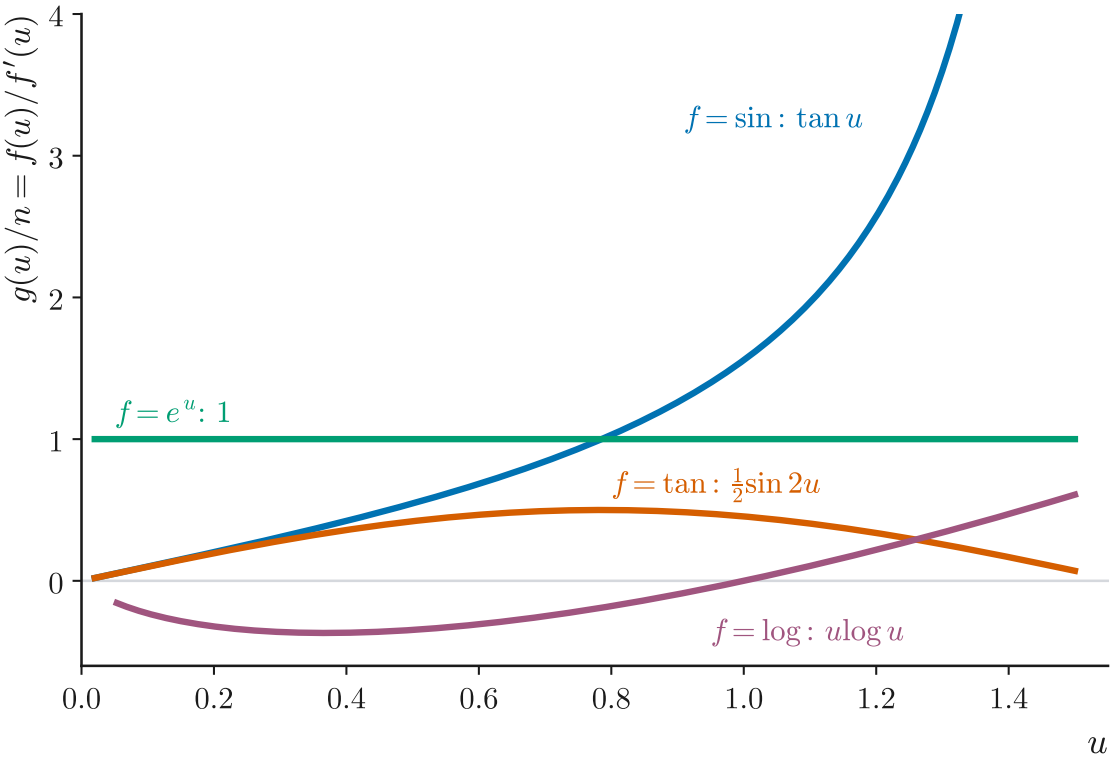


Figure G28.1. The extension function $f(u)/f'(u)$ for each choice of f. ESA ★★

ADDED FOR CLARITY S006·13 · S006·15 · S006·17 · ESA May-24 1a

How to read it. Multiply the curve by the degree n to get $g(u) = xu_x + yu_y$. For $f = \sin$ the sum grows without bound as $u \to \pi/2$; for $f = \tan$ it stays between $\pm\frac{n}{2}$; for $f = e^u$ it is the constant n, so the second order is $-n$; for $f = \log$ it is $nu\log u$, negative for $u < 1$ (on slide 006·12 $u = e^{(\cdots)} > 1$, so it is positive).

M6

Class 007 · Taylor's and Maclaurin's Expansion

29 Why Taylor's and Maclaurin's series?

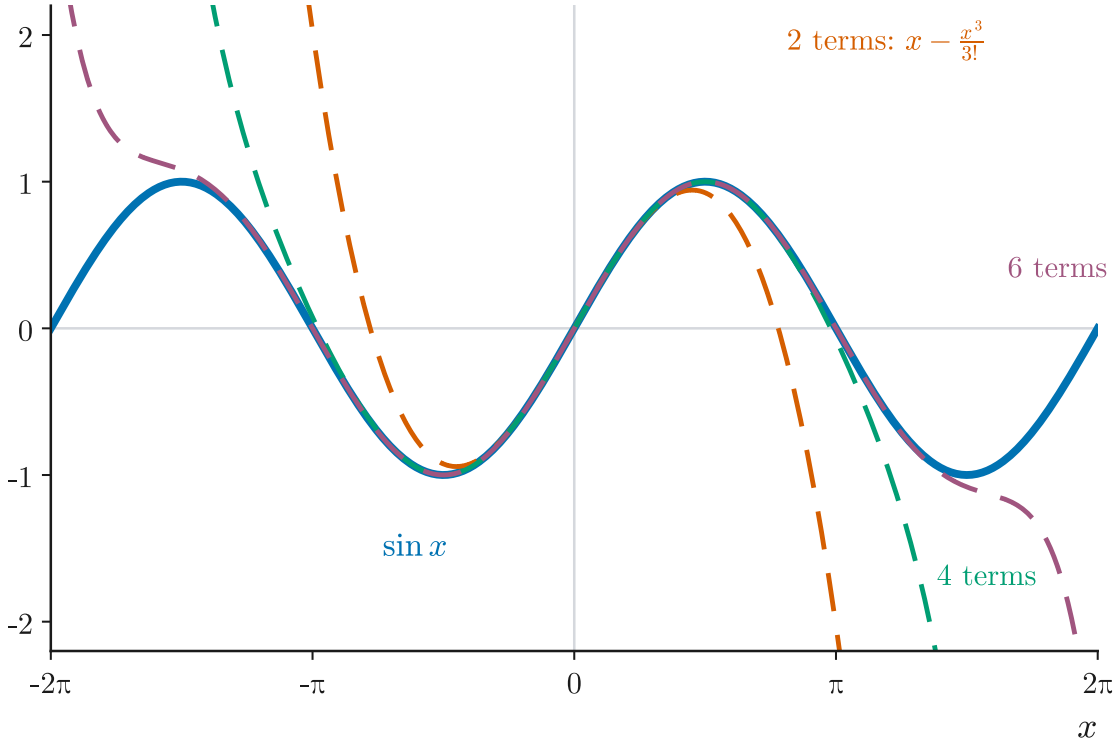


Figure G29.1. $\sin x$ and its Maclaurin polynomials. ESA ★

S007·8 · T007 00:14:45 · T007 00:15:58

How to read it. Blue is $\sin x$; the dashed curves keep 2, 4 and 6 terms of $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots$. Each extra term extends the range where the polynomial follows the function: near 0 all of them agree; far from 0 the short polynomials shoot off. The slide shows 4, 6 and 10 terms; with 10 terms the curves overlap on this range.

30 Taylor's and Maclaurin's series in one variable

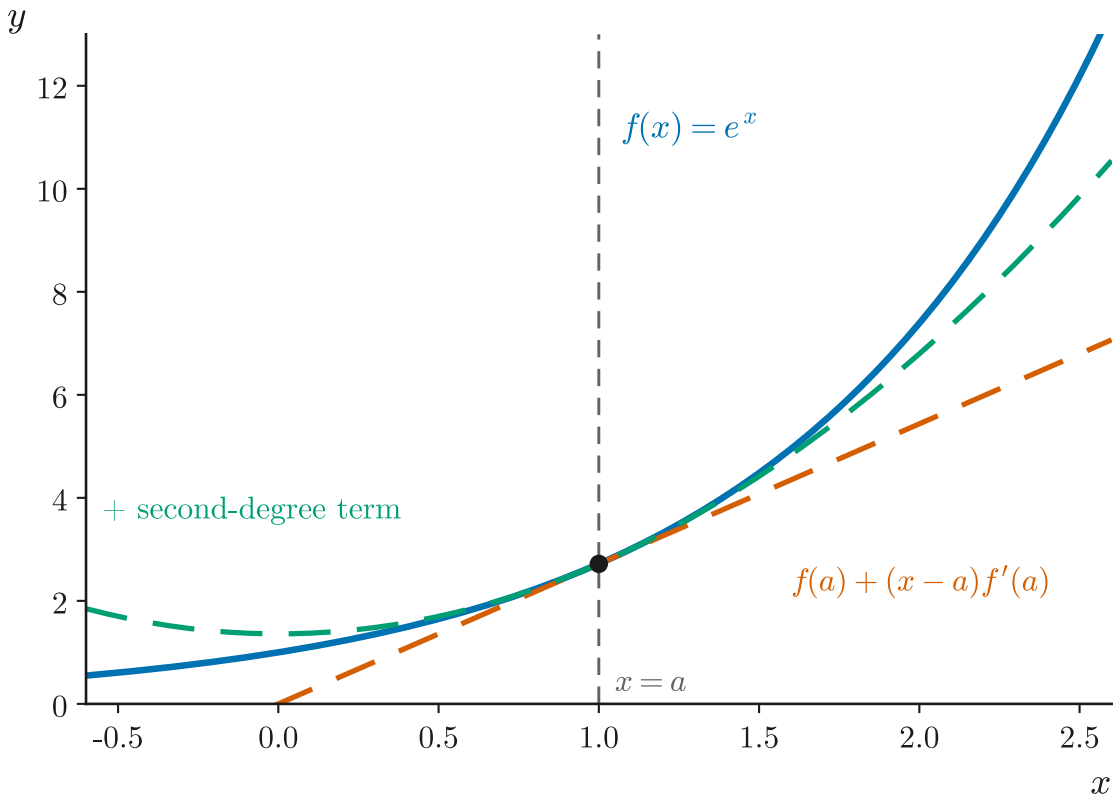


Figure G30.1. The first terms of Taylor's series: tangent line and parabola. ESA ★

ADDED FOR CLARITY T007 00:05:26 · T007 00:05:43

How to read it. $f(x) = e^x$ about $a = 1$. Two terms, $f(a) + (x-a)f'(a)$, give the tangent line (vermilion). Adding $\frac{(x-a)^2}{2!}f''(a)$ gives a parabola (green) that bends with the curve and stays close over a wider range.

31 Taylor's expansion of a function of two variables

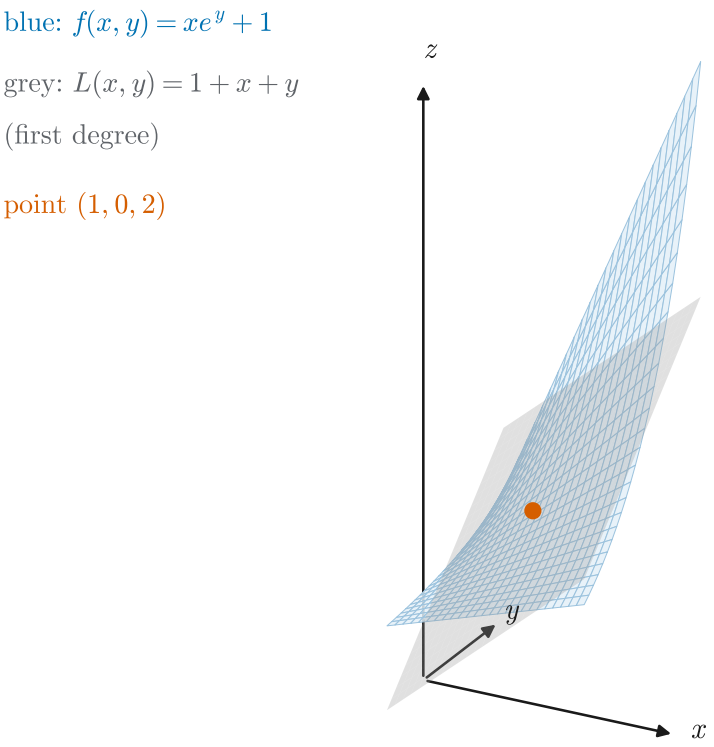


Figure G31.1. $xe^y + 1$ and its first-degree Taylor polynomial. ESA ★

S007·11 · T007 00:20:13

How to read it. The surface $z = xe^y + 1$ (blue mesh) and the plane $z = 1 + x + y$ (grey) touch at $(1, 0, 2)$. The first-degree Taylor polynomial is the tangent plane: it matches the surface's height and slopes at the point and separates from it away from the point.

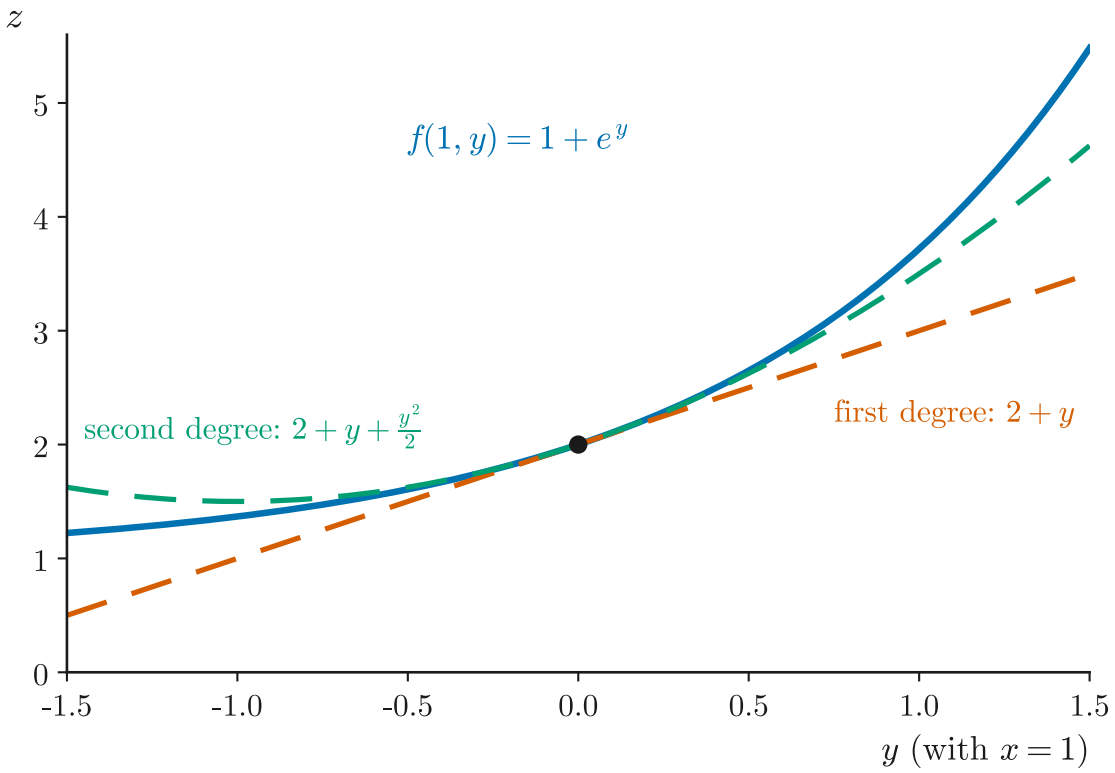


Figure G31.2. Along the line $x = 1$: first- vs second-degree polynomial. ESA ★

ADDED FOR CLARITY S007·11 · T007 00:20:51

How to read it. A slice of Figure G31.1 at $x = 1$: $f(1, y) = 1 + e^y$ (blue), the first-degree polynomial $2 + y$ (vermilion) and the second-degree $1 + x + xy + \frac{y^2}{2} = 2 + y + \frac{y^2}{2}$ (green). The second degree adds curvature and stays close much longer.

M7

Class 008 · Problems on Taylor's and Maclaurin's Series

33 Taylor expansion about a point (a, b)

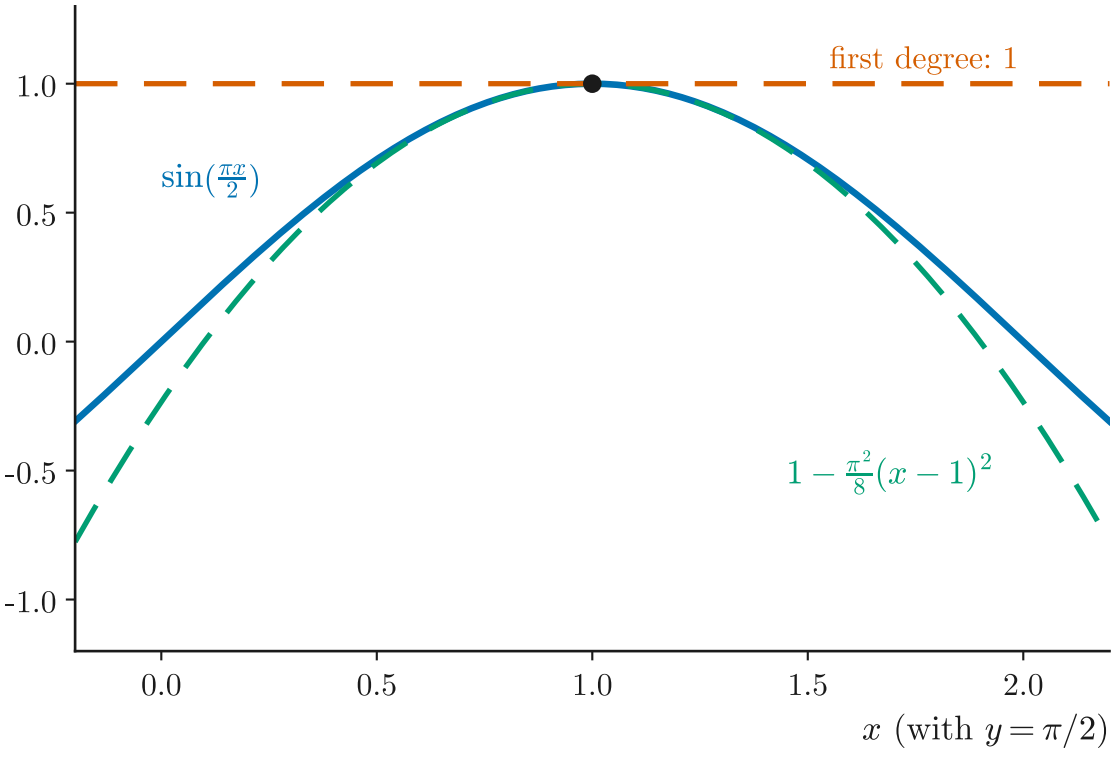


Figure G33.1. $\sin(xy)$ near $(1, \pi/2)$: a slice at $y = \pi/2$. ESA ★

ADDED FOR CLARITY S008·14 · S008·16

How to read it. Along $y = \frac{\pi}{2}$, $\sin(xy) = \sin\frac{\pi x}{2}$ (blue) has a maximum at $x = 1$, so both first partials vanish and the first-degree polynomial is the flat line 1 (vermilion). The second-degree polynomial $1 - \frac{\pi^2}{8}(x-1)^2$ (green) follows the curvature near $x = 1$.

34 Maclaurin expansion about the origin

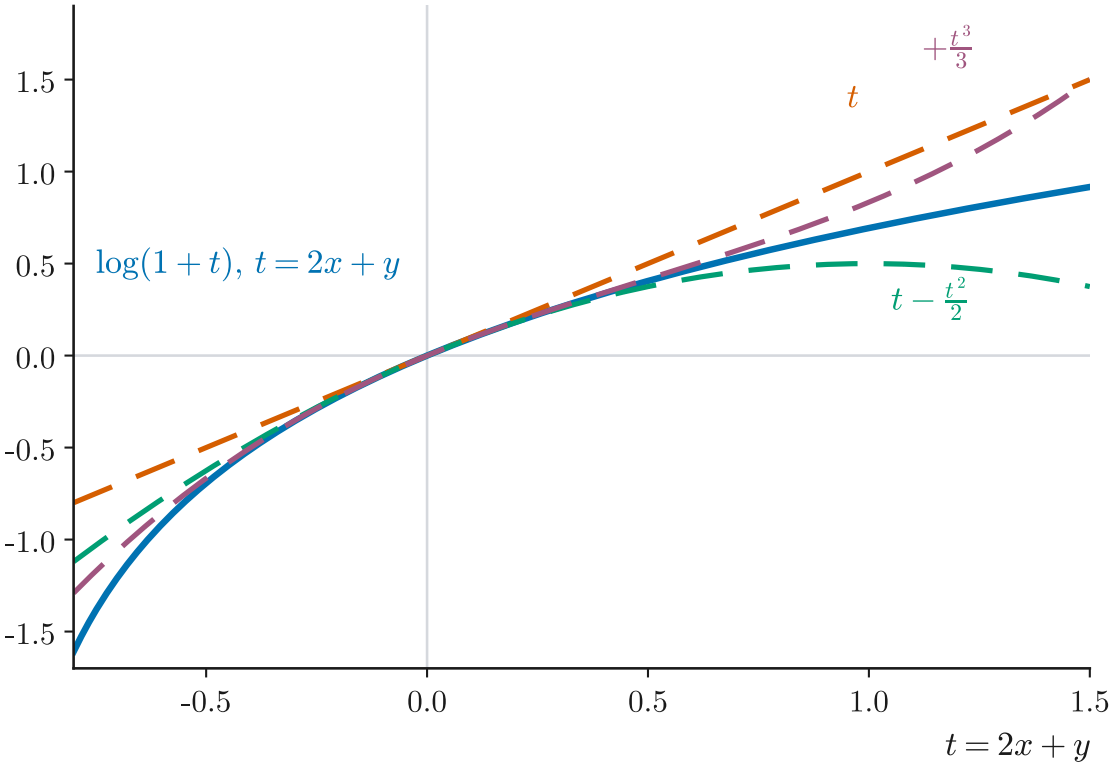


Figure G34.1. $\log(2x + y + 1)$ as a function of $t = 2x + y$. ESA ★★

ADDED FOR CLARITY S008·5 · S008·6

How to read it. The function depends only on $t = 2x + y$, so its Maclaurin polynomials are t, $t - \frac{t^2}{2}$ and $t - \frac{t^2}{2} + \frac{t^3}{3}$. Each extra degree follows $\log(1 + t)$ (blue) further from $t = 0$; all fail as $t \to -1$, where the logarithm blows up.

35 Approximate values from a Taylor polynomial

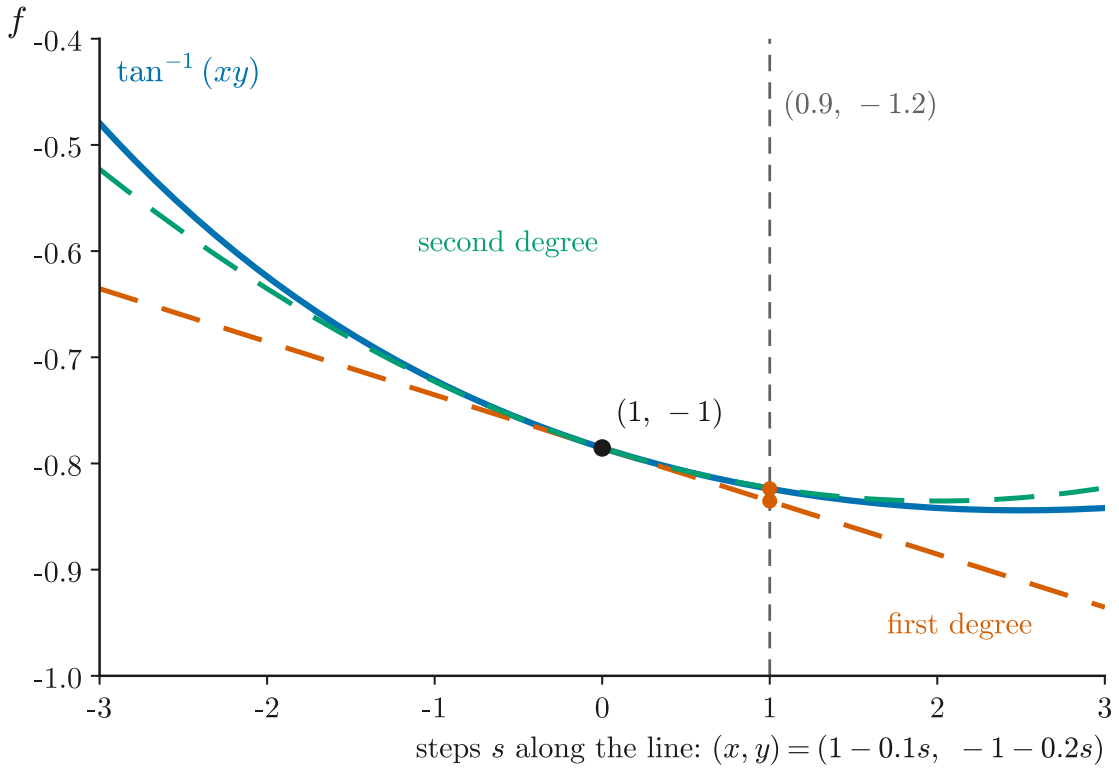


Figure G35.1. Approximating $\tan^{-1}(xy)$ at $(0.9, -1.2)$. ESA ★★

ADDED FOR CLARITY S008·7 · S008·9 · T008 00:14:18

How to read it. Values along the straight line from $(1, -1)$ (black dot, $s = 0$) through $(0.9, -1.2)$ ($s = 1$, grey dashed line). At $s = 1$ the first-degree polynomial (vermilion) gives -0.835, while the function (blue) is -0.824; the second degree (green) is almost on the curve there.

M8

Class 009 · Maxima and Minima of a Function of Two Variables

36 Maxima, minima and saddle points

two local maxima (peaks)

vermilion: horizontal tangent plane

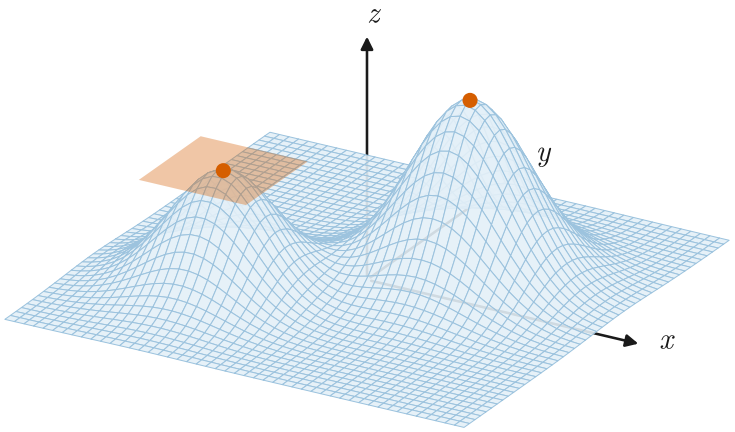


Figure G36.1. Local maxima: peaks with horizontal tangent planes. ESA ★

S009·3 · T009 00:00:16

How to read it. A surface with two peaks (illustrative: a sum of two bell shapes). Both peaks are local maxima; the taller one is also the largest value overall. The vermilion square is the horizontal tangent plane at the lower peak.

$z = x^2 + y^2$

minimum at $(0,0)$: $rt - s^2 = 4 > 0,\ r = 2 > 0$

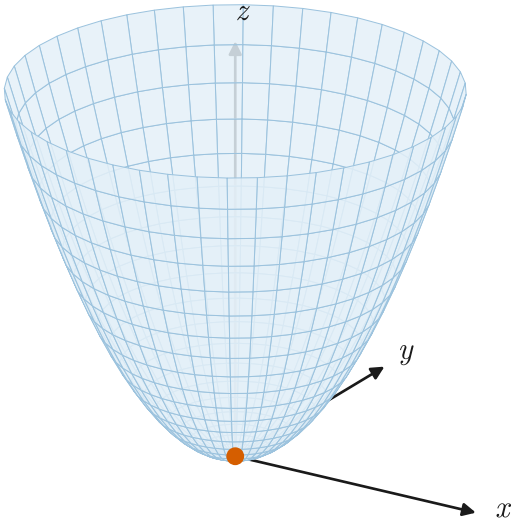


Figure G36.2. A local minimum: $z = x^2 + y^2$. ESA ★

S009·4 · T009 00:02:11

How to read it. The bowl $z = x^2 + y^2$ has its lowest point at the origin. Every nearby point is higher: $\Delta > 0$. Its test values are $r = t = 2$, $s = 0$, $rt - s^2 = 4 > 0$, $r > 0$ (W38.1).

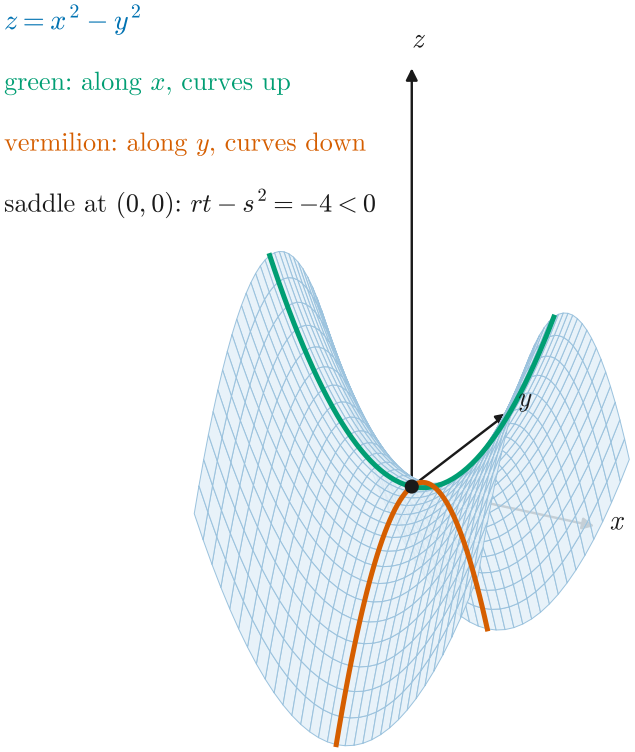


Figure G36.3. A saddle point: $z = x^2 - y^2$. ESA ★★

S009·5 · S009·7

How to read it. Along the x-direction (green) the surface curves up, along the y-direction (vermilion) it curves down. The origin is stationary but neither a maximum nor a minimum; $rt - s^2 = -4 < 0$ (W38.2).

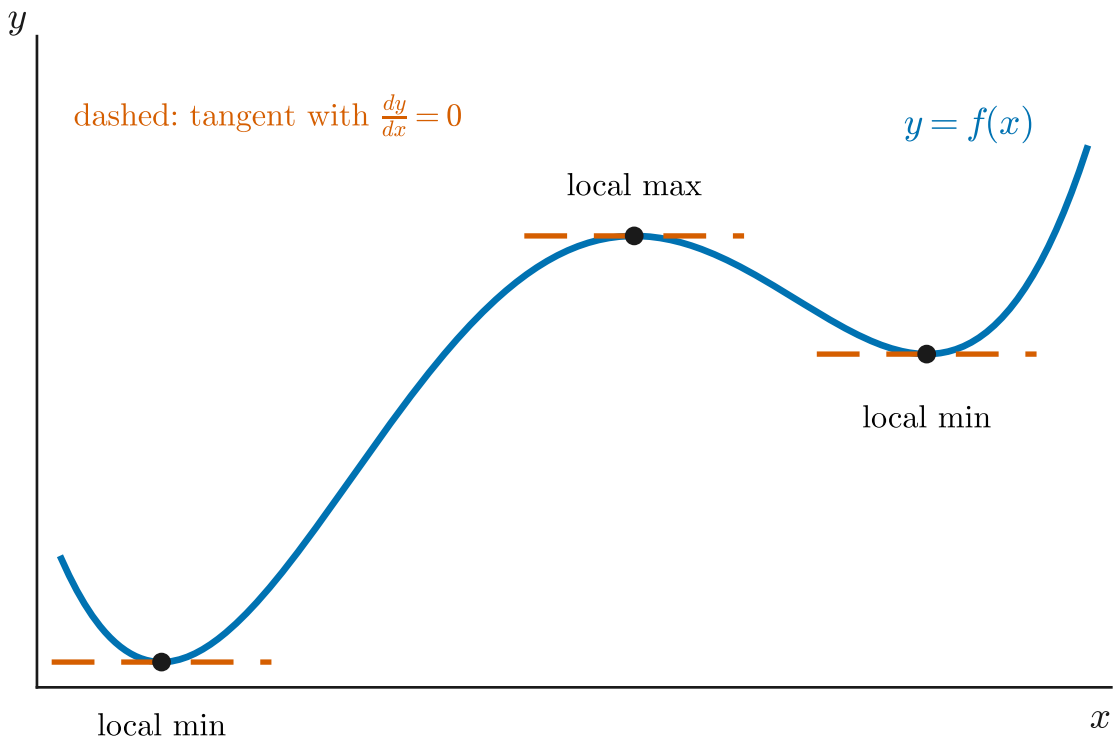


Figure G36.4. One-variable picture: maxima, minima and $\frac{dy}{dx} = 0$. ESA ★

T009 00:00:50 · T009 00:01:10

How to read it. On $y = f(x)$ the tangent is horizontal (slope 0) at every local maximum and minimum (short dashed segments). The two-variable test copies this: $f_x = f_y = 0$ replaces $\frac{dy}{dx} = 0$, and $rt - s^2$ with r replaces $\frac{d^2y}{dx^2}$.

M9

Class 010 · Problems on Maxima and Minima

39 The method, and the ESA cubic functions

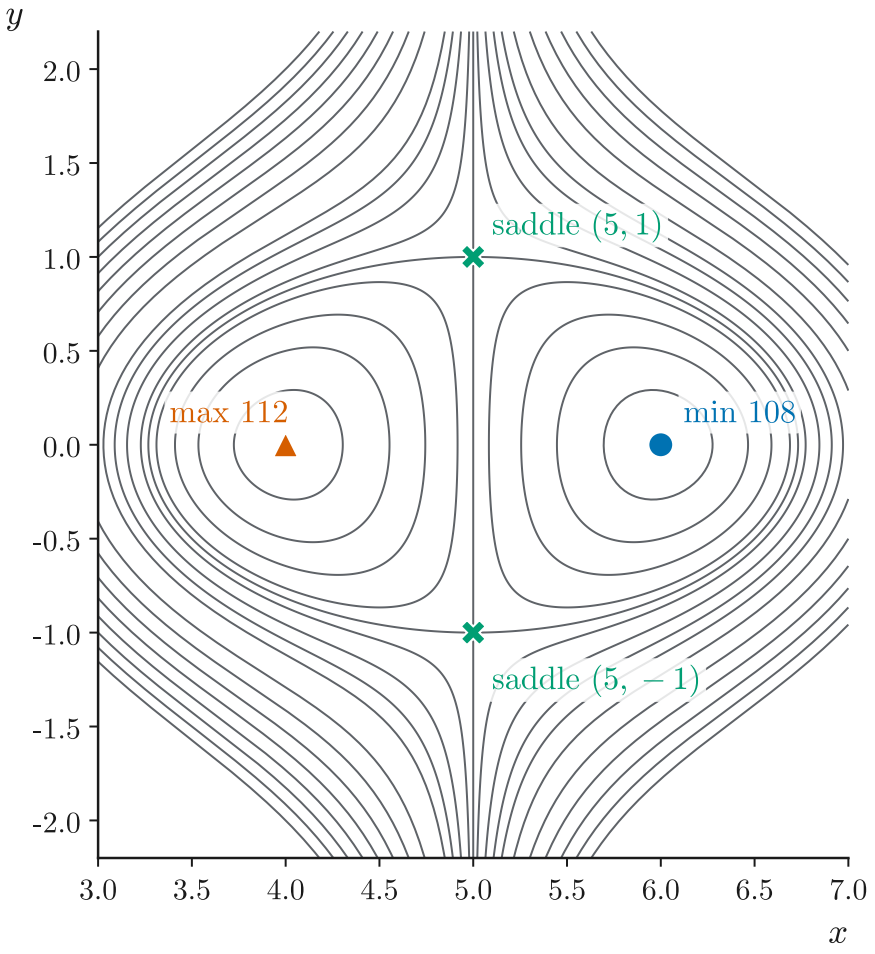


Figure G39.1. Level curves of $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$. ESA ★★★

ADDED FOR CLARITY S010·8 · ESA Dec-23 1c · ESA Jan-24 s1 1a

How to read it. Grey lines are level curves $f =$ constant. Around the maximum $(4, 0)$ and the minimum $(6, 0)$ they close into small loops; at the saddle points $(5, \pm 1)$ two level curves cross, the sign of the picture of a saddle.

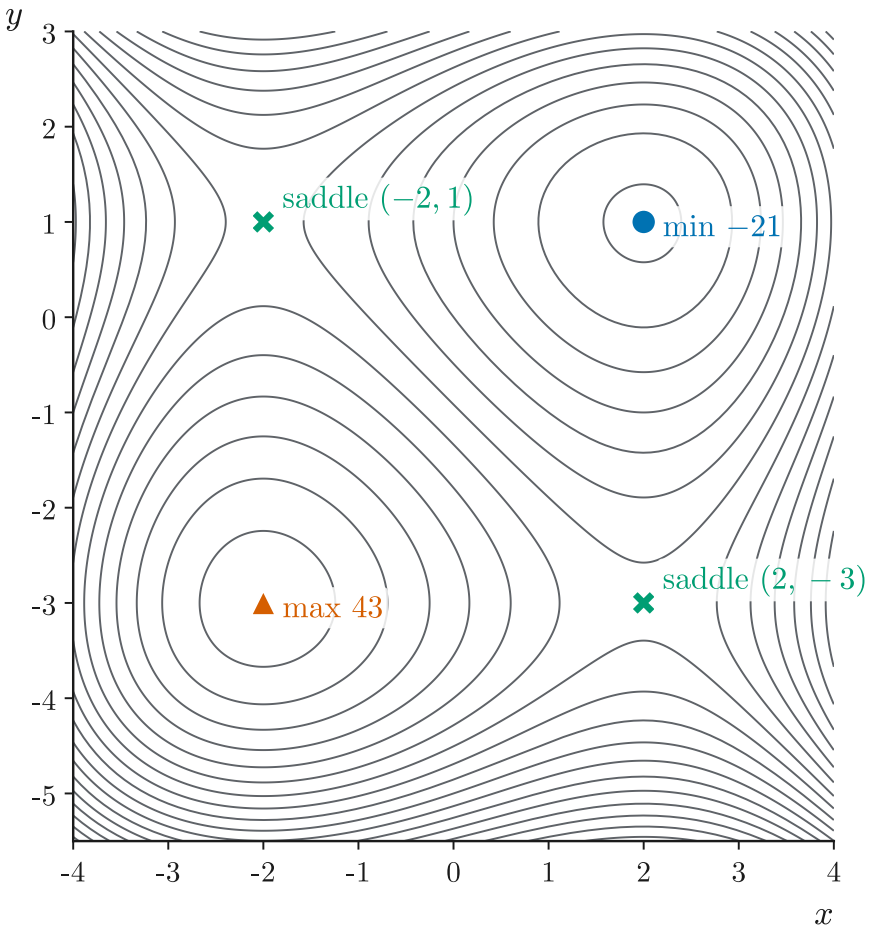


Figure G39.2. Level curves of $x^3 - 12x + y^3 + 3y^2 - 9y$. ESA ★★★

ADDED FOR CLARITY ESA Jan-24 s2 1a · CW U1 P29

How to read it. The four stationary points sit at the corners of a rectangle, $x = \pm 2$, $y = 1, -3$. Closed loops surround the maximum $(-2, -3)$ and the minimum $(2, 1)$; curves cross at the saddles $(2, -3)$ and $(-2, 1)$.

40 Functions with many stationary points

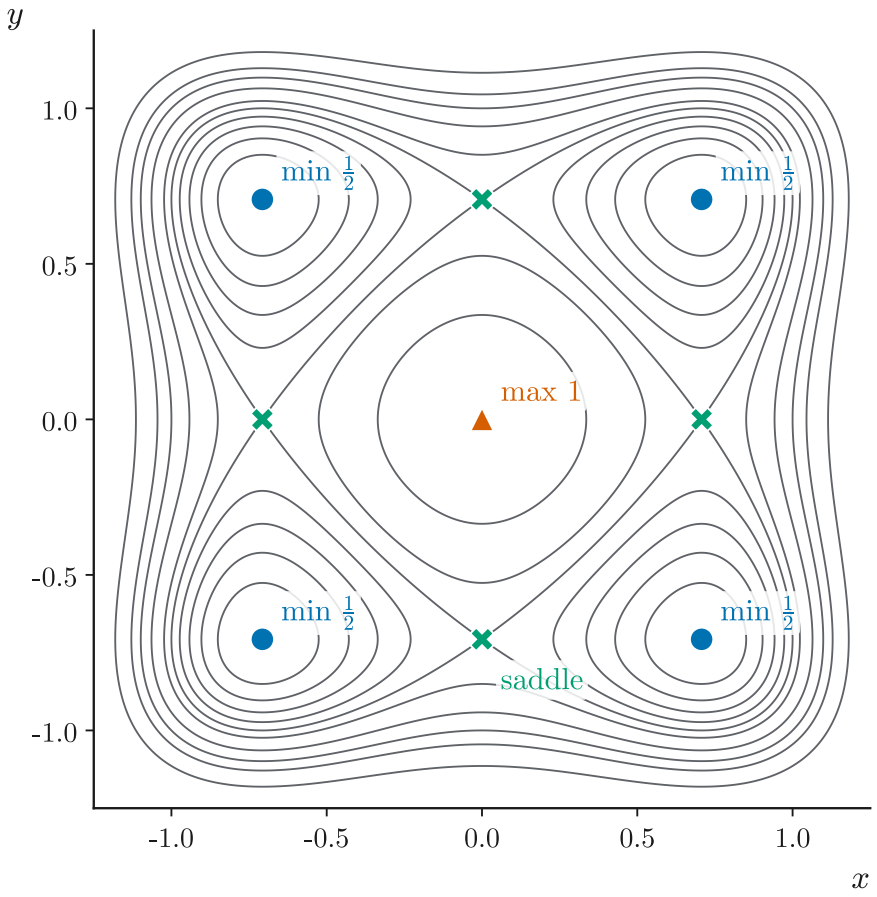


Figure G40.1. Level curves of $x^4 + y^4 - x^2 - y^2 + 1$. ESA ★★

ADDED FOR CLARITY S010·11 · S010·12

How to read it. Nine stationary points on a $\mathbf{3 \times 3}$ grid: a maximum at the centre, minima at the four corners $(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}})$ and saddles at the four edge midpoints on the axes.

41 Applied problems: eliminate one variable first

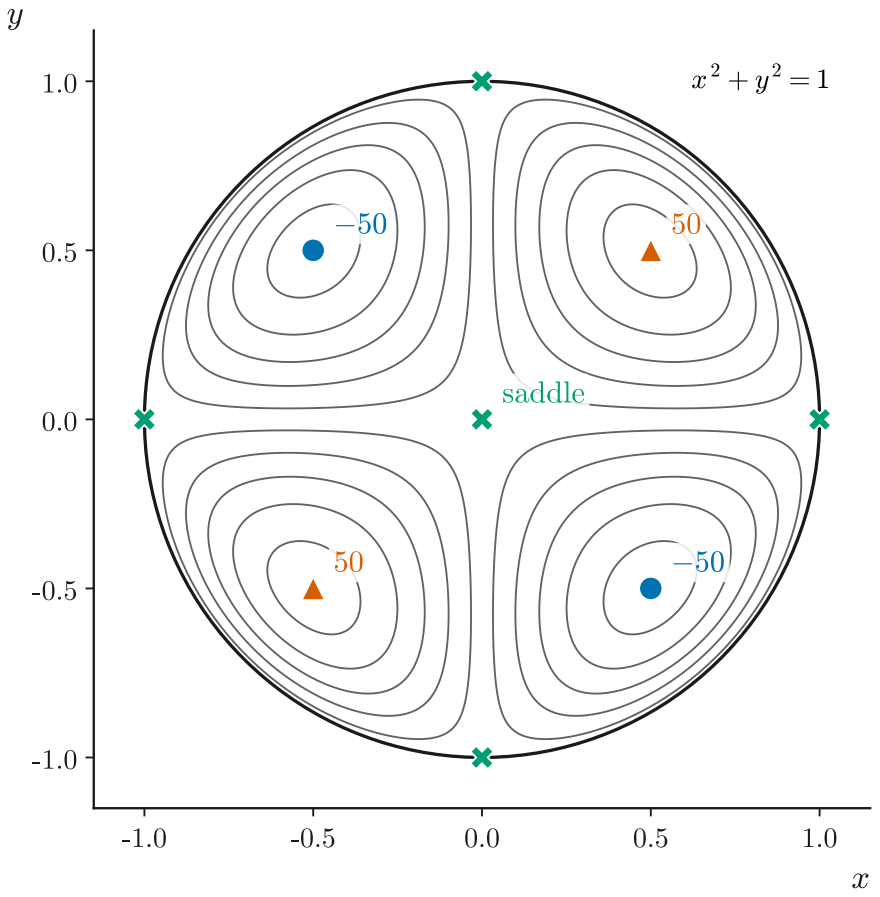


Figure G41.1. Temperature $400xy(1 - x^2 - y^2)$ over the unit disk. ESA ★★

ADDED FOR CLARITY S010·3 · S010·4

How to read it. Inside the unit circle (the projection of the sphere) the temperature is positive in the first and third quadrants and negative in the others. It peaks at $\mathbf{50}$ at $(\frac{1}{2}, \frac{1}{2})$ and $(-\frac{1}{2}, -\frac{1}{2})$ and is lowest, $-\mathbf{50}$, at the other two. The origin and $(\pm 1, 0)$, $(0, \pm 1)$, where $T = 0$, are saddles.

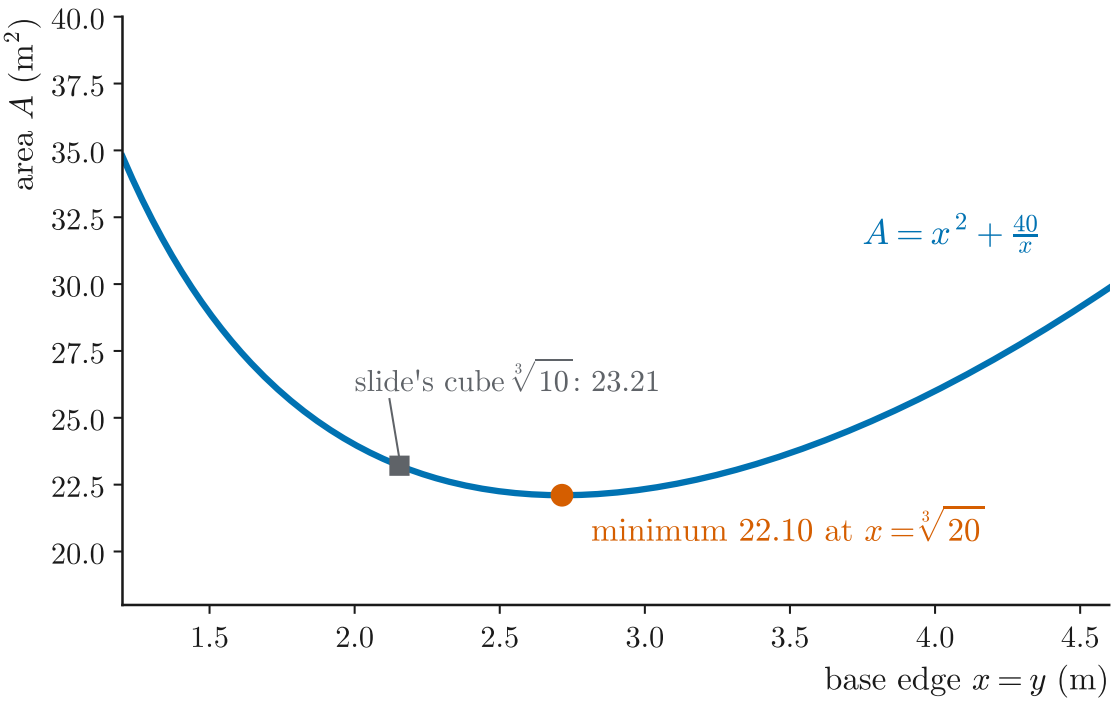


Figure G41.2. Metal used by an open box of 10 m³ with a square base. ESA ★

ADDED FOR CLARITY S010·6 · CW U1 P34

How to read it. Along $x = y$ the area is $A = x^2 + \frac{40}{x}$ (blue). Its minimum, **22.10** m², is at $x = \sqrt[3]{20} \approx 2.71$ m. The slide's cube, $x = \sqrt[3]{10} \approx 2.15$ m, sits higher on the curve (**23.21** m²).

M10

Class 011 · Lagrange's Method of Undetermined Multipliers

42 Lagrange's method: the auxiliary function and the procedure

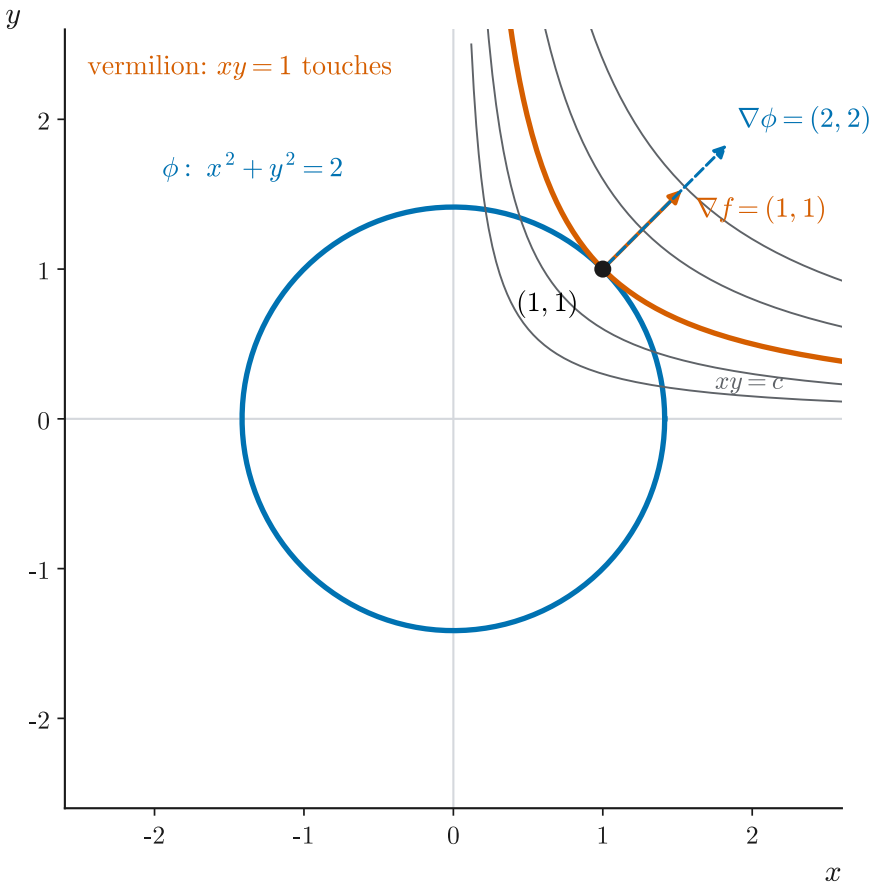


Figure G42.1. Why the method works: a level curve touches the constraint. ESA ★

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How to read it. Maximise $f = xy$ on the circle $x^2 + y^2 = 2$ (blue). The level curves $xy = c$ (grey hyperbolas) cross the circle until $c = 1$, where $xy = 1$ (vermilion) just touches it at $(1, 1)$. There ∇f and $\nabla\phi$ point the same way (arrows): $\nabla f + \lambda\nabla\phi = 0$ with $\lambda = -\frac{1}{2}$.

43 Geometric problems: inscribed solids, nearest and farthest points

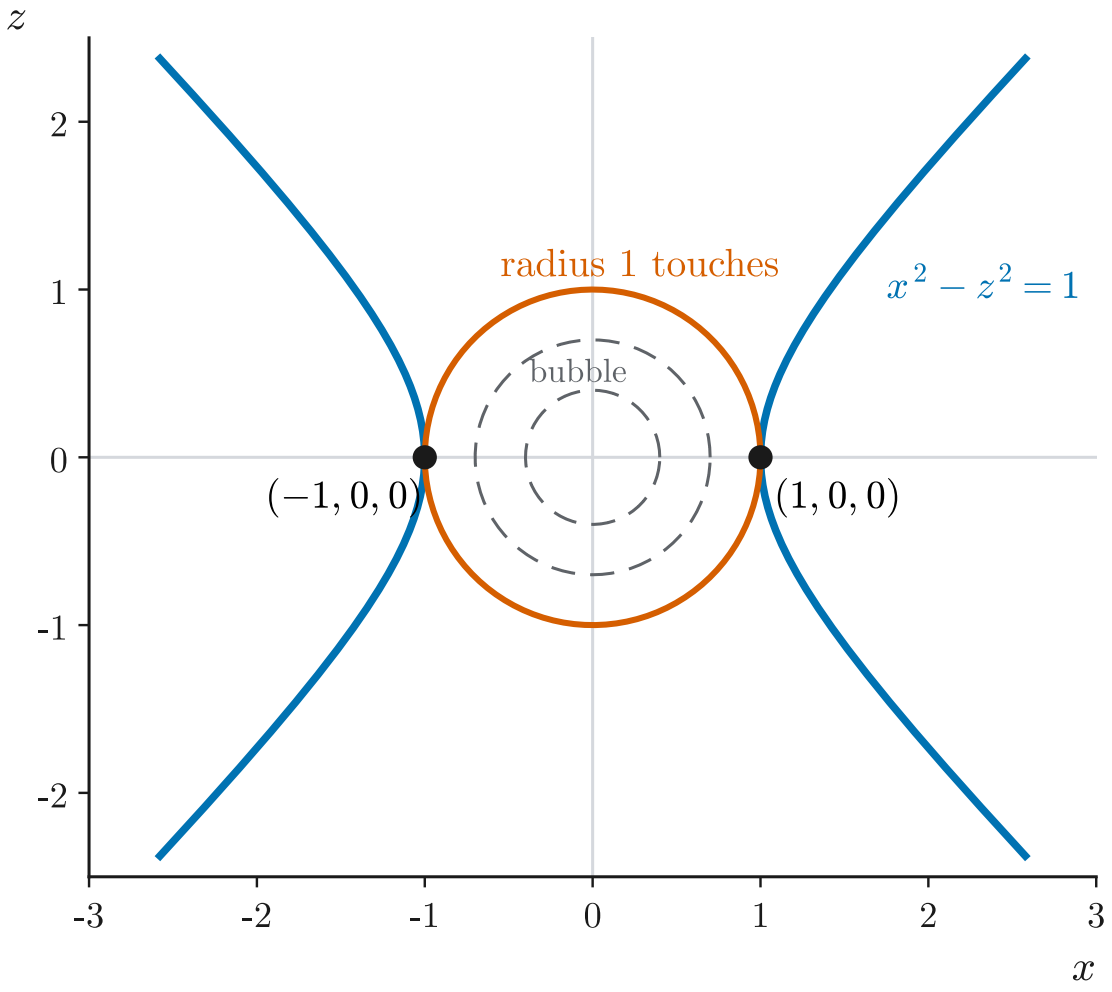


Figure G43.1. The soap bubble: $x^2 - z^2 = 1$ nearest the origin. ESA ★★

S011·7 · S011·9

How to read it. Cross-section $y = 0$. The hyperbolic cylinder appears as the hyperbola $x^2 - z^2 = 1$ (blue). Circles centred at the origin (grey, growing like a soap bubble) first touch it at $(\pm 1, 0)$ with radius 1 (vermilion): the nearest points $(\pm 1, 0, 0)$.

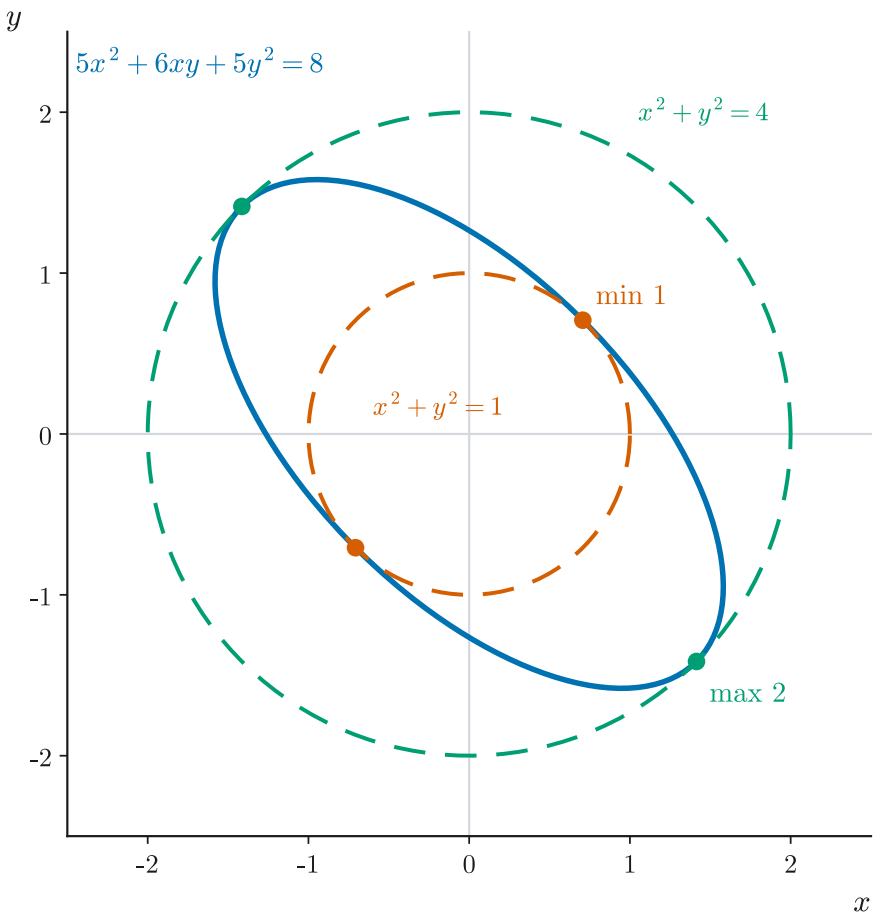


Figure G43.2. $5x^2 + 6xy + 5y^2 = 8$ between the circles of radius 1 and 2. ESA ★★★

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How to read it. The curve is an ellipse (blue) tilted at $45°$. The circle $x^2 + y^2 = 1$ (vermilion) touches it from inside at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$: minimum distance 1. The circle $x^2 + y^2 = 4$ (green) touches it from outside at $\pm(\sqrt{2}, -\sqrt{2})$: maximum distance 2.